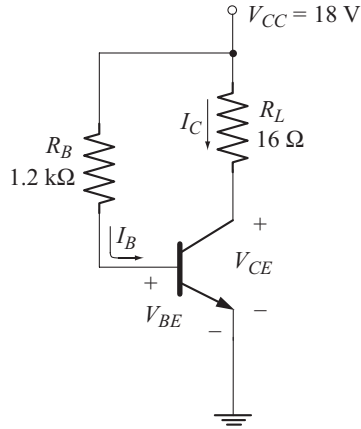


Solution

(a) The dc equivalent circuit is shown below.



$$I_B = I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{18 \text{ V} - 0.7 \text{ V}}{1.2 \text{ k}\Omega} = 14.42 \text{ mA}$$

$$I_C = I_{CQ} = \beta I_{BQ} = 40 \times 14.42 \text{ mA} = 576.8 \text{ mA}$$

Applying KVL to the collector circuit.

$$\begin{aligned} V_{CC} &= I_C R_L + V_{CE} \\ \therefore V_{CE} &= V_{CEQ} = V_{CC} - I_{CQ} R_L \\ &= 18 \text{ V} - (576.8 \text{ mA})(16 \Omega) = 8.77 \text{ V} \end{aligned}$$

DC input power

$$\begin{aligned} P_{i(dc)} &= V_{CC} \times I_{CQ} \\ &= 18 \text{ V} \times 576.8 \text{ mA} \\ P_{i(dc)} &= 10.4 \text{ W} \end{aligned} \quad (\text{A})$$

rms base current is given as

$$I_{B(\text{rms})} = 5 \text{ mA}$$

$\therefore$ rms collector current is

$$I_{C(\text{rms})} = \beta I_{B(\text{rms})} = (40)(5 \text{ mA}) = 200 \text{ mA}$$

Peak collector current or peak load current is

$$I_{C(p)} = \sqrt{2} \times \text{rms collector current}$$

$$I_{C(p)} = \sqrt{2} I_{C(\text{rms})} = \sqrt{2} \times 200 \text{ mA}$$

$$I_{C(p)} = 282.8 \text{ mA}$$

ac power delivered to the load is

$$P_{o(ac)} = \frac{I_{C(p)}^2 R_L}{2} = \frac{(282.8 \text{ mA})^2 (16 \Omega)}{2} = 0.64 \text{ W} \quad (\text{B})$$

Conversion efficiency

$$\% \eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100\% = \frac{0.64 \text{ W}}{10.4 \text{ W}} \times 100\% = 6.15\%$$

(b) The dc power dissipated in the load

$$P_{L(dc)} = I_{CQ}^2 R_L = (576.8 \text{ mA})^2 \times 16 \Omega = 5.3 \text{ W} \quad (\text{C})$$

Power dissipated in the collector

$$P_C = V_{CEQ} I_{CQ} = (8.77 \text{ V}) (576.8 \text{ mA}) = 5.1 \text{ W} \quad (\text{D})$$

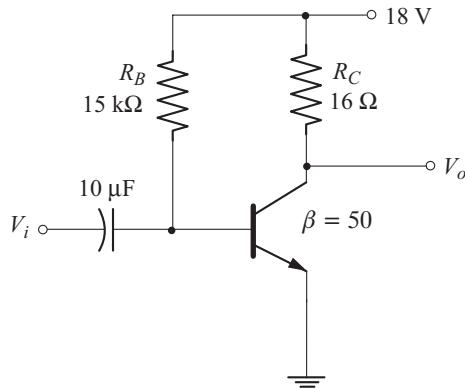
From equations (A), (C) and (D), observe that

$$P_{i(dc)} = P_{L(dc)} + P_C$$

Example 7.18

In the circuit shown below, the input signal results in a peak base current of 1 mA

- Calculate the ac output power
- Calculate the dc input power dissipated by the circuit
- Calculate the efficiency.



Solution

$$\begin{aligned} \text{(a)} \quad I_{B(p)} &= 1 \text{ mA} \\ I_{C(p)} &= \beta I_{B(p)} = (50) (1 \text{ mA}) = 50 \text{ mA} \\ P_{o(ac)} &= \frac{I_{C(p)}^2 R_C}{2} = \frac{(50 \text{ mA})^2}{2} (16 \Omega) = 20 \text{ mW} \end{aligned}$$

$$\begin{aligned}
 (b) \quad I_{BQ} &= \frac{V_{CC} - V_{BE}}{R_B} = \frac{18\text{ V} - 0.7\text{ V}}{15\text{ k}\Omega} = 1.153\text{ mA} \\
 I_{CQ} &= \beta I_{BQ} = (50)(1.153\text{ mA}) = 57.65\text{ mA} \\
 P_{i(dc)} &= V_{CC} I_{CQ} = (18\text{ V})(57.65\text{ mA}) = 1.037\text{ W} \\
 (c) \quad \eta &= \frac{P_{o(ac)}}{P_{i(dc)}} = \frac{20\text{ mW}}{1.037\text{ W}} = 0.0193 \text{ or } 1.93\%
 \end{aligned}$$

Example 7.19

For the circuit of previous example,

- calculate the maximum output power
- if R_B is adjusted so that the Q point lies at the centre of the DC load line, calculate the input power for a maximum output power of 1.5 W. What is the efficiency in this case?

Solution

$$\begin{aligned}
 (a) \quad P_{o(ac)} &= \frac{V_{CE(p)}^2}{2R_C} = \frac{I_{C(p)}^2}{2} R_C \\
 I_{CQ} &= 57.65\text{ mA} \quad \text{as before.}
 \end{aligned}$$

To obtain maximum undistorted output, $I_{C(p)}$ should not exceed I_{CQ}

$$\text{i.e.} \quad I_{C(p)} = I_{CQ} = 57.65\text{ mA}$$

Maximum ac output power is

$$P_{o(ac)} = \frac{I_{CQ}^2}{2} R_C = \frac{(57.65\text{ mA})^2}{2} (16\text{ }\Omega) = 26.59\text{ mW}$$

(b) Since the Q point is at the centre

$$\begin{aligned}
 V_{CEQ} &= \frac{V_{CC}}{2} = \frac{18\text{ V}}{2} = 9\text{ V} \\
 I_{CQ} &= \frac{V_{CC} - V_{CEQ}}{R_C} \quad [\text{From KVL equation of collector circuit}] \\
 &= \frac{18\text{ V} - 9\text{ V}}{16\text{ }\Omega} = 0.5625\text{ A} \\
 P_{i(dc)} &= V_{CC} I_{CQ} = (18\text{ V})(0.5625\text{ A}) = 10.125\text{ W} \\
 \eta &= \frac{P_{o(ac)}}{P_{i(dc)}} = \frac{1.5\text{ W}}{10.125\text{ W}} = 0.1482 \text{ or } 14.82\%
 \end{aligned}$$

Note:

For the biasing condition given in part (b), the ac power delivered to the load is maximum when

$$V_{CE(p)} = V_{CEQ} \text{ and } I_{C(p)} = I_{CQ}$$

Under this condition the maximum ac output power is

$$P_{o(ac)} = \frac{V_{CE(p)} I_{C(p)}}{2} = \frac{(9\text{ V})(0.5625\text{ A})}{2} = 2.53\text{ W}$$

The corresponding efficiency is,

$$\eta = \frac{2.53\text{ W}}{10.125\text{ W}} \approx 0.25 \text{ or } 25\%$$

which is equal to the ideal value.

Therefore to obtain maximum efficiency when the Q point is at the centre of the dc load line, the input signal strength should be large enough to result in

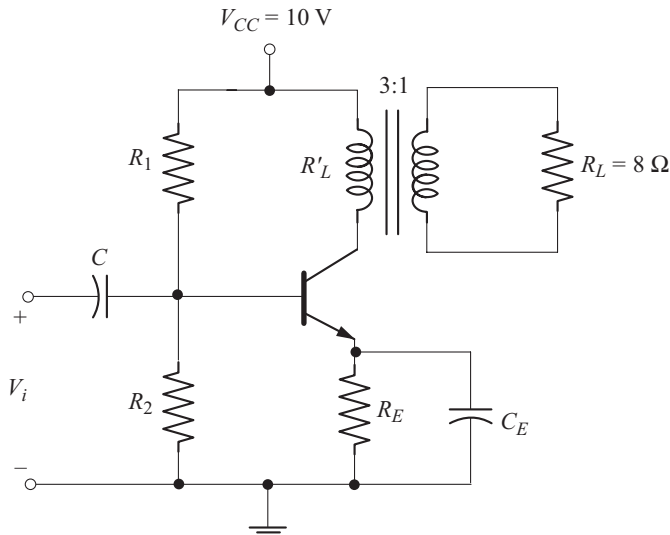
$$V_{CE(p)} = V_{CEQ} \text{ and } I_{C(p)} = I_{CQ}$$

Otherwise the efficiency will be smaller than 25% even if the Q point is at the centre of the dc load line.

Example 7.20

For the circuit shown, the dc base current is 5 mA and the ac input signal results in a peak base current swing of 4 mA. Assume silicon transistor with $\beta = 30$. Find

- ac power delivered to the load
- dc power drawn by the circuit
- conversion efficiency



Solution

Given,

$$I_{BQ} = 5 \text{ mA} \quad I_{B(p)} = 4 \text{ mA} \quad \beta = 30 \quad \frac{N_1}{N_2} = 3$$

For transformer-coupled class A power amplifier

$$V_{CEQ} = V_{CC} = 10 \text{ V} \quad (\text{see Fig. 7.12})$$

$$I_{CQ} = \beta I_{BQ} = 30 \times 5 \text{ mA} = 150 \text{ mA}$$

Peak collector current

$$I_{C(p)} = \beta I_{B(p)} = 30 \times 4 \text{ mA} = 120 \text{ mA}$$

(a) From Equation (7.40), ac power delivered to the load is

$$P_{o(ac)} = \frac{V_{CE(p)}^2}{2R'_L} = \frac{I_{C(p)}^2}{2} R'_L$$

$$\text{But} \quad R'_L = R_L \left(\frac{N_1}{N_2} \right)^2 = (8 \Omega) (3)^2 = 72 \Omega$$

$$\therefore P_{o(ac)} = \frac{(120 \text{ mA})^2 (72 \Omega)}{2} = 0.52 \text{ W}$$

(b) dc power drawn by the circuit is

$$P_{i(dc)} = V_{CC} I_{CQ} = (10 \text{ V}) (150 \text{ mA}) = 1.5 \text{ W}$$

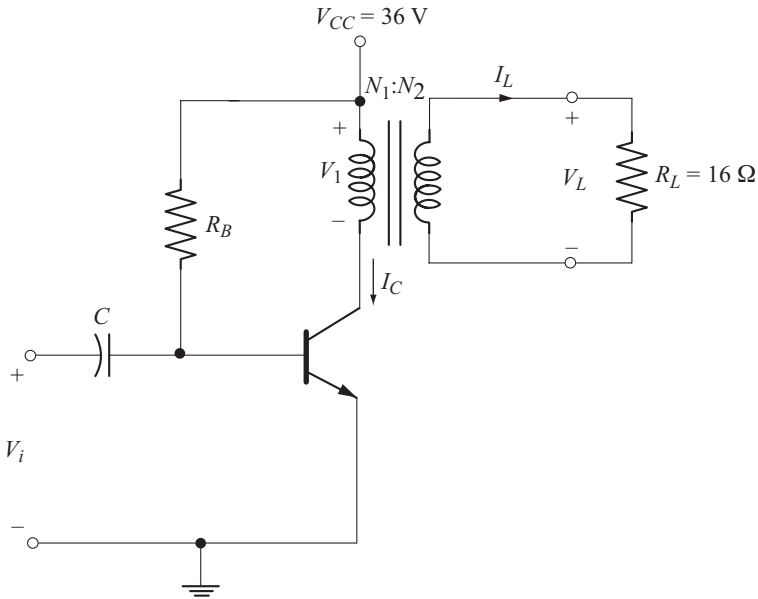
(c) Conversion efficiency

$$\begin{aligned} \% \eta &= \frac{P_{o(ac)}}{P_{i(dc)}} \times 100\% \\ &= \frac{0.52 \text{ W}}{1.5 \text{ W}} \times 100\% = 34.7\% \end{aligned}$$

Example 7.21

A transformer-coupled class A amplifier drives a 16Ω loud speaker through a 4:1 transformer. With $V_{CC} = 36 \text{ V}$ the circuit delivers 2 W to the load. Find,

- power across the transformer primary,
- rms voltage across the load,
- rms voltage across the transformer primary,
- rms values of load current and primary current, and
- conversion efficiency if the dc collector current is 150 mA.

Solution

$$\frac{V_1}{V_L} = \frac{N_1}{N_2} = \frac{I_L}{I_C} \quad (\text{A})$$

Given,

$$\frac{N_1}{N_2} = 4 \quad R_L = 16 \, \Omega \quad V_{CC} = 36 \, \text{V} \quad I_{CQ} = 150 \, \text{mA}$$

Power delivered to the load R_L is, $P_L = 2 \, \text{W}$

(a) Power across transformer primary.

Assuming 100 % transformer efficiency, the power across transformer primary is

$$P_{pri} = P_{o(ac)} = P_L = 2 \, \text{W} \quad (\text{B})$$

(b) rms voltage across the load

$$P_L = \frac{V_{L(\text{rms})}^2}{R_L}$$

$$\therefore V_{L(\text{rms})} = \sqrt{P_L R_L} = \sqrt{(2 \, \text{W})(16 \, \Omega)} = 5.65 \, \text{V}$$

(c) rms voltage across the transformer primary.

From Equation (A)

$$\frac{V_{1(\text{rms})}}{V_{L(\text{rms})}} = \frac{N_1}{N_2} = 4$$

$$\therefore V_{1(\text{rms})} = 4 V_{L(\text{rms})} = (4)(5.65 \, \text{V}) = 22.6 \, \text{V}$$

(d) rms load and primary current

$$P_L = I_{L(\text{rms})}^2 \cdot R_L$$

$$\therefore I_{L(\text{rms})} = \sqrt{\frac{P_L}{R_L}} = \sqrt{\frac{2\text{ W}}{16\Omega}} = 353.55\text{ mA}$$

From Equation (A)

$$\frac{I_{L(\text{rms})}}{I_{C(\text{rms})}} = \frac{N_1}{N_2} = 4$$

$$\therefore I_{C(\text{rms})} = \frac{I_{L(\text{rms})}}{4} = \frac{353.55\text{ mA}}{4} = 88.39\text{ mA}$$

(e) Conversion efficiency

$$\% \eta = \frac{P_{o(\text{ac})}}{P_{i(\text{dc})}} \times 100\% \quad (\text{C})$$

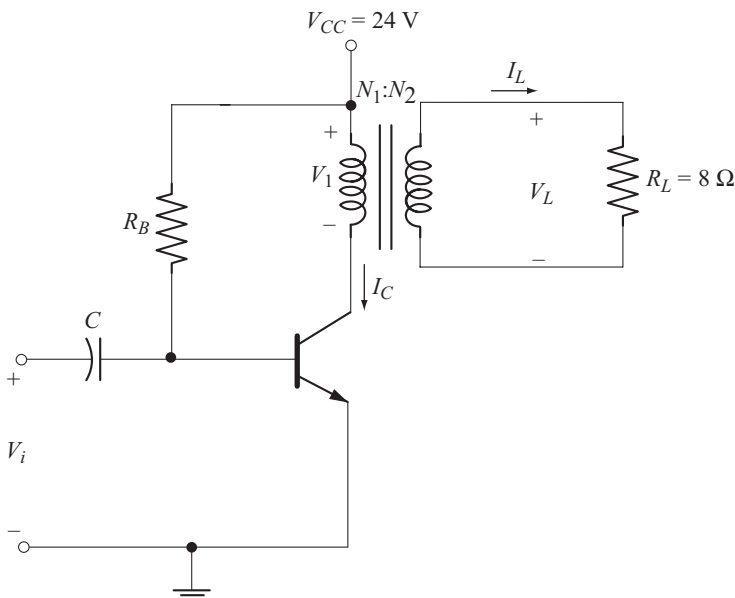
$$P_{i(\text{dc})} = V_{CC} I_{CQ} = 36\text{ V} \times 150\text{ mA} = 5.4\text{ W}$$

$$\therefore \text{From (B) and (C)} \quad \% \eta = \frac{2\text{ W}}{5.4\text{ W}} \times 100 = 37.03\%$$

Example 7.22

A transformer-coupled class A amplifier drives a load of 8Ω through a 3:1 transformer. With $V_{CC} = 24\text{ V}$, the circuit delivers 2 W to the load. The transformer efficiency is 80%. Find,

- power across the transformer primary,
- rms voltage across load and transformer primary
- rms values of load current and primary current
- conversion efficiency if dc collector current is 260 mA .



Solution

$$\frac{V_1}{V_L} = \frac{N_1}{N_2} = \frac{I_L}{I_C} \quad (A)$$

Given, $\frac{N_1}{N_2} = 3 \quad \eta_{TFR} = 0.8 \text{ or } 80\% \quad R_L = 8 \Omega$

$$V_{CC} = 24 \text{ V} \quad I_{CQ} = 260 \text{ mA} \quad P_L = 2 \text{ W}$$

(a) $\eta_{TFR} = \frac{P_L}{P_{o(ac)}}$

$$P_{pri} = P_{o(ac)} = \frac{P_L}{0.8} = \frac{2 \text{ W}}{0.8} = 2.5 \text{ W}$$

(b) $V_{L(rms)} = \sqrt{P_L R_L} = \sqrt{(2 \text{ W})(8 \Omega)} = 4 \text{ V}$

(c) $\frac{V_{1(rms)}}{V_{L(rms)}} = \frac{N_1}{N_2} = 3$

$$\therefore V_{1(rms)} = 3 V_{L(rms)} = 3 \times 4 \text{ V} = 12 \text{ V}$$

(d) rms load current

$$I_{L(rms)} = \sqrt{\frac{P_L}{R_L}} = \sqrt{\frac{2 \text{ W}}{8 \Omega}} = 0.5 \text{ A}$$

$$\frac{I_{L(rms)}}{I_{C(rms)}} = \frac{N_1}{N_2} = 3$$

rms primary current

$$I_{C(rms)} = \frac{I_{L(rms)}}{3} = \frac{0.5 \text{ A}}{3} = 0.1667 \text{ A}$$

(e) Conversion efficiency

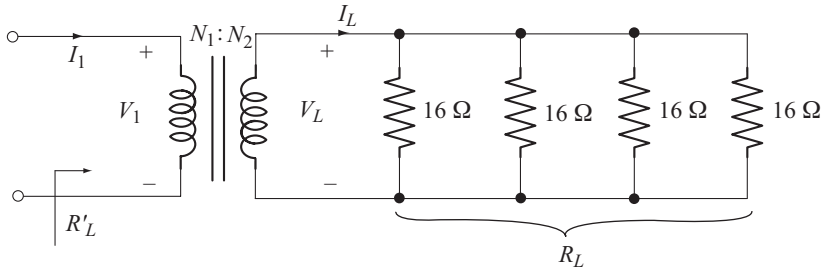
$$\% \eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100\%$$

$$P_{i(dc)} = V_{CC} \times I_{CQ} = 24 \text{ V} \times 260 \text{ mA} = 6.24 \text{ W}$$

$$\therefore \% \eta = \frac{2.5 \text{ W}}{6.24 \text{ W}} \times 100\% = 40\%$$

Example 7.23

Find the turns ratio of the transformer required to connect four parallel 16Ω loud speakers so that they appear as an $8 \text{ k}\Omega$ effective load.

**Solution**

$$R_L = (16\ \Omega \parallel 16\ \Omega \parallel 16\ \Omega \parallel 16\ \Omega) = \frac{16\ \Omega}{4} = 4\ \Omega$$

Given, $R'_L = 8\ \text{k}\Omega$

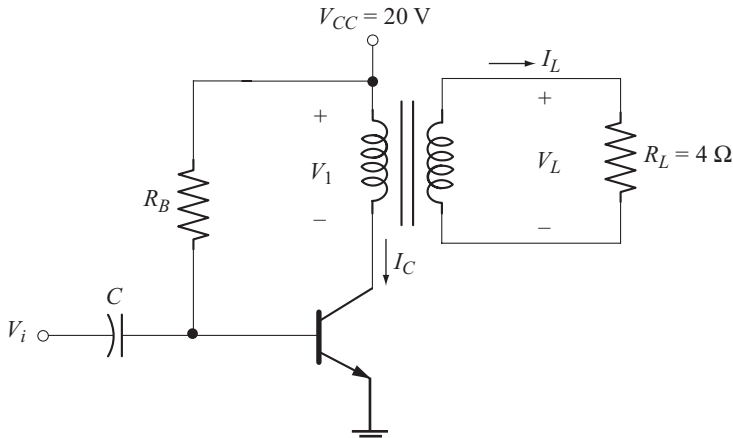
$$R'_L = \left(\frac{N_1}{N_2} \right)^2 R_L$$

$$\therefore \frac{N_1}{N_2} = \sqrt{\frac{R'_L}{R_L}} = \sqrt{\frac{8000\ \Omega}{4\ \Omega}} = \sqrt{2000} = 44.7$$

Example 7.24

A transformer-coupled class A power amplifier is required to deliver a maximum of 5 W to a 4 Ω load. The quiescent point is adjusted for symmetrical clipping and the collector supply voltage is $V_{CC} = 20\ \text{V}$. Assuming ideal characteristics and taking $V_{CE(\min)} = 0$, find

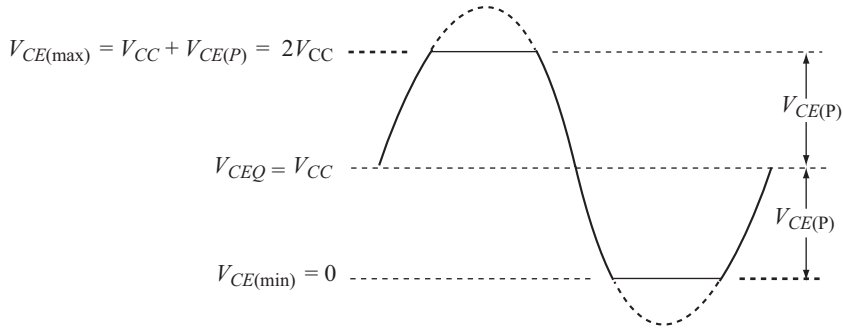
- the transformer turns ratio,
- peak collector current,
- quiescent operating point I_{CQ} , V_{CEQ} , and
- collector circuit efficiency.



Solution

Given, $V_{CC} = 20 \text{ V}$ $R_L = 4 \Omega$ $P_L = 5 \text{ W}$ $V_{CE(\min)} = 0$

The output voltage waveform is shown below.



From the figure

$$V_{CE(p)} = V_{CC} = 20 \text{ V}$$

(a) Assuming ideal transformer

$$P_{o(ac)} = P_L = 5 \text{ W}$$

But from Equation (7.40)

$$P_{o(ac)} = \frac{V_{CE(p)}^2}{2R'_L}$$

$$\therefore R'_L = \frac{V_{CE(p)}^2}{2P_{o(ac)}} = \frac{(20 \text{ V})^2}{2 \times 5 \text{ W}} = 40 \Omega$$

$$R'_L = \left(\frac{N_1}{N_2} \right)^2 \times R_L$$

$$\Rightarrow \left(\frac{N_1}{N_2} \right) = \sqrt{\frac{R'_L}{R_L}} = \sqrt{\frac{40 \Omega}{4 \Omega}} = \sqrt{10}$$

$$\therefore N_1 : N_2 = \sqrt{10} : 1$$

(b) Peak collector current

$$P_{o(ac)} = I_{C(rms)}^2 R'_L$$

$$I_{C(rms)}^2 = \frac{P_{o(ac)}}{R'_L} = \frac{5 \text{ W}}{40 \Omega} = 0.125 \text{ A}^2$$

Peak collector current

$$I_{C(p)} = \sqrt{2} I_{C(rms)} = \sqrt{2} \times 0.353 \text{ A} = 0.5 \text{ A}$$

(c) Quiescent operating point

$$V_{CEQ} = V_{CC} = 20 \text{ V}$$

$$I_{CQ} = I_{C(p)} = 0.5$$

$\therefore$ Q -point is at (20 V, 0.5 A)

(d) Collector circuit efficiency

$$\% \eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100 \%$$

$$P_{i(dc)} = V_{CC} I_{CQ} = 20 \text{ V} \times 0.5 \text{ A} = 10 \text{ W}$$

$$\therefore \% \eta = \frac{5 \text{ W}}{10 \text{ W}} \times 100 \% = 50 \%$$

Observe that since we have assumed ideal conditions the conversion efficiency is 50% as expected.

Example 7.25

The following data is available for a class A transformer-coupled power amplifier.

$$\begin{array}{ll} V_{CE(max)} = 18.5 \text{ volts} & I_{C(max)} = 250 \text{ mA} \\ V_{CE(min)} = 1.5 \text{ volts} & I_{C(min)} = 25 \text{ mA} \\ V_{CC} = 10 \text{ volts} & I_{CQ} = 140 \text{ mA} \\ R_L = 8 \Omega \end{array}$$

Find:

- AC power delivered to the load
- Conversion efficiency
- Transformer turns ratio

Solution

(a) For a class A transformer-coupled power amplifier,

$$V_{CE(max)} = V_{CC} + V_{CE(p)} \quad (A)$$

$$V_{CE(min)} = V_{CC} - V_{CE(p)} \quad (B)$$

Solving for $V_{CE(p)}$ we get

$$V_{CE(p)} = \frac{V_{CE(max)} - V_{CE(min)}}{2} = \frac{18.5 \text{ V} - 1.5 \text{ V}}{2} = 8.5 \text{ V}$$

Similarly

$$I_{C(p)} = \frac{I_{C(max)} - I_{C(min)}}{2} = \frac{250 \text{ mA} - 25 \text{ mA}}{2} = 112.5 \text{ mA}$$

AC power delivered to the load

$$\begin{aligned} P_{o(ac)} &= \frac{V_{CE(p)} I_{C(p)}}{2} \\ &= \frac{(8.5 \text{ V})(112.5 \text{ mA})}{2} = 0.478 \text{ W} \end{aligned}$$

$$(b) \quad \% \eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100 \%$$

$$\begin{aligned} P_{i(dc)} &= V_{CC} \times I_{CQ} \\ &= (10 \text{ V})(140 \text{ mA}) = 1.4 \text{ W} \end{aligned}$$

$$\therefore \% \eta = \frac{0.478 \text{ W}}{1.4 \text{ W}} \times 100 \% = 34.14\%$$

$$\begin{aligned} (c) \quad V_{CE(p)} &= I_{C(p)} R'_L \\ R'_L &= \frac{V_{CE(p)}}{I_{C(p)}} = \frac{8.5 \text{ V}}{112.5 \text{ mA}} = 75.56 \Omega \\ R'_L &= \left(\frac{N_1}{N_2} \right)^2 R_L \\ \frac{N_1}{N_2} &= \sqrt{\frac{R'_L}{R_L}} = \sqrt{\frac{75.56 \Omega}{8 \Omega}} = 3.07 \end{aligned}$$

Example 7.26

Calculate the efficiency of transformer-coupled class A power amplifier for a collector dc supply of 15 V and outputs of

$$(a) \quad V_{CE(p)} = 3.75 \text{ V}$$

$$(b) \quad V_{CE(p)} = 7.5 \text{ V}$$

$$(c) \quad V_{CE(p)} = 15 \text{ V}$$

Comment on the result.

Solution

$$\text{Given} \quad V_{CC} = 15 \text{ V}$$

The conversion efficiency of transformer-coupled class A power amplifier is given by

$$\% \eta = 50 \left[\frac{V_{CE(\max)} - V_{CE(\min)}}{V_{CE(\max)} + V_{CE(\min)}} \right] \% \quad (A)$$

$$\text{where} \quad V_{CE(\max)} = V_{CC} + V_{CE(p)} \quad (B)$$

$$V_{CE(\min)} = V_{CC} - V_{CE(p)} \quad (C)$$

(a) $V_{CE(p)} = 3.75$ volts

$$\begin{aligned} V_{CE(\max)} &= V_{CC} + V_{CE(p)} \\ &= 15 \text{ V} + 3.75 \text{ V} \\ &= 18.75 \text{ V} \end{aligned}$$

$$\begin{aligned} V_{CE(\min)} &= V_{CC} - V_{CE(p)} \\ &= 15 \text{ V} - 3.75 \text{ V} \\ &= 11.25 \text{ V} \end{aligned}$$

$$\begin{aligned} \% \eta &= 50 \times \frac{18.75 \text{ V} - 11.25 \text{ V}}{18.75 \text{ V} + 11.25 \text{ V}} \\ &= 12.5\% \end{aligned}$$

(b) $V_{CE(p)} = 7.5$ V

$$\begin{aligned} V_{CE(\max)} &= 15 \text{ V} + 7.5 \text{ V} \\ &= 22.5 \text{ V} \end{aligned}$$

$$\begin{aligned} V_{CE(\min)} &= 15 \text{ V} - 7.5 \text{ V} \\ &= 7.5 \text{ V} \end{aligned}$$

$$\% \eta = 50 \times \frac{22.5 \text{ V} - 7.5 \text{ V}}{22.5 \text{ V} + 7.5 \text{ V}} = 25\%$$

(c) $V_{CE(p)} = 15$ V

$$V_{CE(\max)} = 15 \text{ V} + 15 \text{ V} = 30 \text{ V}$$

$$V_{CE(\min)} = 15 \text{ V} - 15 \text{ V} = 0 \text{ V}$$

$$\% \eta = 50 \times \frac{30 \text{ V} - 0}{30 \text{ V} + 0} = 50\%$$

From the above calculations we find that, efficiency increases with increase in $V_{CE(p)}$ and a maximum efficiency of 50% is obtained when, $V_{CE(p)} = V_{CC} = 15$ V.

◆ 7.6 CLASS B PUSH-PULL POWER AMPLIFIER

As mentioned in Section 7.1.2 in class B power amplifier, the Q -point is located at cutoff. This means that the base-emitter voltage is at zero volts.

As a result the transistor conducts current for only one-half cycle of the signal cycle. To obtain output for the full cycle of the signal it is necessary to use two transistors, one conducting during the positive half cycle and the other during the negative half cycle as shown in Fig. 7.15. The configuration of Fig. 7.15(a) requires two dc power supplies where as a single dc power supply is sufficient for the configuration of Fig. 7.15(b).

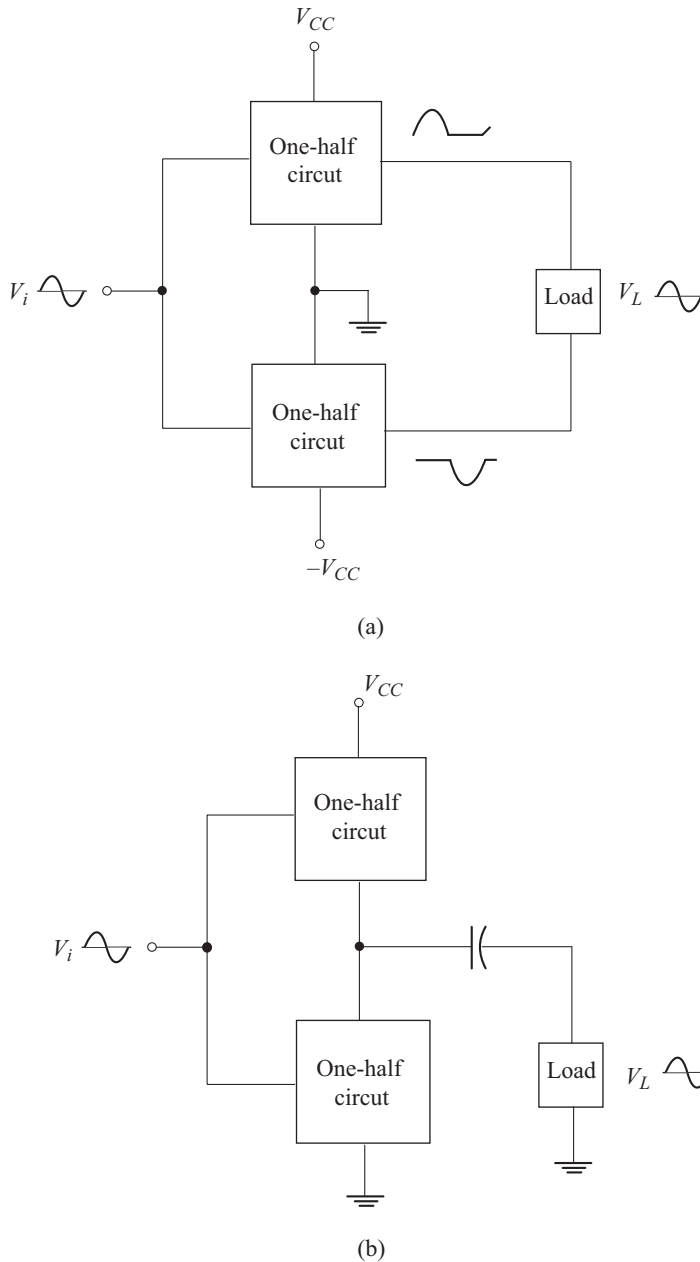


Fig. 7.15 Push pull amplifier
 (a) Using two dc power supplies (b) Using single dc power supply

7.6.1 Operation of Class B Push-pull Power Amplifier

Figure 7.16 shows the circuit of class B push-pull power amplifier. Push-pull configuration is used to eliminate harmonic distortion introduced by the non-linearity of the dynamic transfer characteristic of the amplifying device.

The input transformer is a 1 : 1 : 1 transformer which is used to provide two equal voltages which are 180° out of phase with each other. The voltage divider network consisting of R_1 and R_2 is used to keep Q_1 and Q_2 in the verge of conduction. Note that the Q point is slightly above cut-off and as a result there will be no cross-over distortion. The output transformer is used to provide impedance matching between the amplifier output and the load.

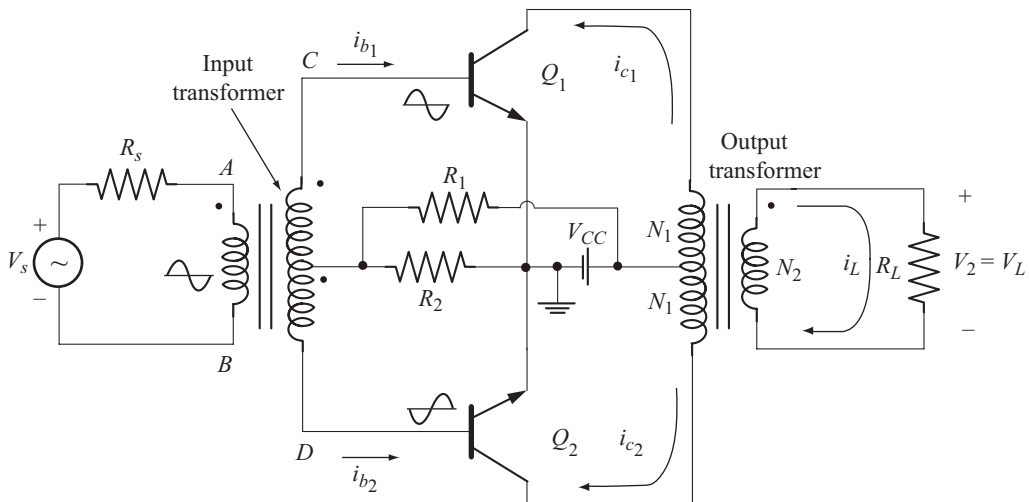


Fig. 7.16 Class B push-pull power amplifier

During the positive half cycle of the input signal v_s , the voltage at C is positive going while the voltage at D is negative going. Since the transistors are biased at cutoff, the base-emitter junction of Q_1 gets forward biased whereas the base-emitter junction of Q_2 goes further below cut-off. Thus i_{c1} increases above zero (note that $I_{CQ} = 0$) while i_{c2} is zero.

$$i_L = \frac{N_1}{N_2} i_{c1} \quad (7.92)$$

It is easy to see that during the negative half cycle of the input signal v_s , Q_1 is *off* and Q_2 conducts.

$$i_L = -\frac{N_1}{N_2} i_{c2} \quad (7.93)$$

The negative sign for i_L is due to the fact that i_{c1} and i_{c2} are in opposite directions.

Combining equations (7.92) and (7.93) we can write

$$i_L = \begin{cases} \frac{N_1}{N_2} i_{c1} & \text{for } 0 \leq \omega t \leq \pi \\ -\frac{N_1}{N_2} i_{c2} & \text{for } \pi \leq \omega t \leq 2\pi \end{cases}$$

$$\text{or} \quad i_L = \frac{N_1}{N_2} [i_{c1} - i_{c2}] \quad (7.94)$$

The input and output voltage and current waveforms are in Fig. 7.17.

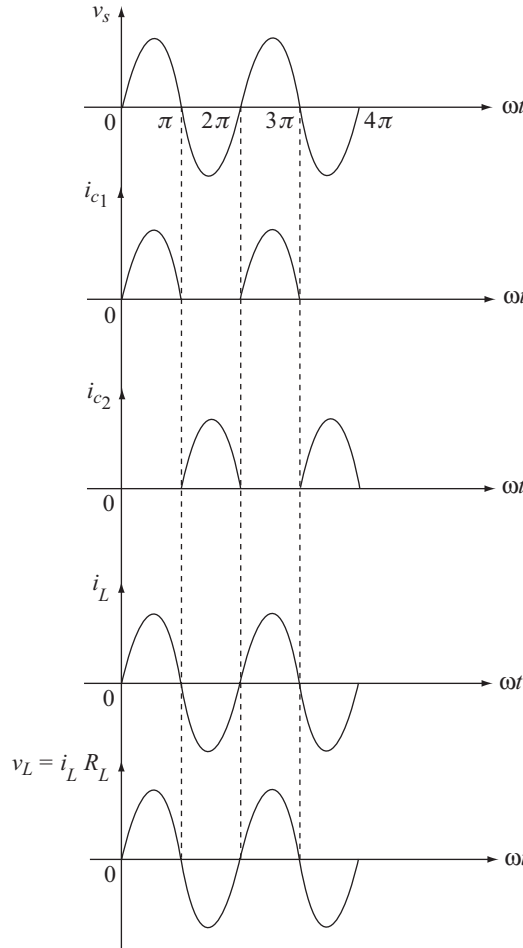


Fig. 7.17 Input and Output waveforms of class B push-pull amplifier

Elimination of Even Harmonic Distortion with Push-pull Operation

From Equation (7.72) total instantaneous collector current of transistor Q_1 is

$$i_{c1} = I_{CQ} + B_0 + B_1 \cos \omega t + B_2 \cos 2\omega t + B_3 \cos 3\omega t + B_4 \cos 4\omega t + \dots \quad (7.95)$$

Observe in the circuit of Fig. 7.16 that i_{c1} and i_{c2} are flowing in opposite directions. Therefore, they differ in phase by 180° .

$$\text{Hence} \quad i_{c2}(\omega t) = i_{c1}(\omega t + \pi) \quad (7.96)$$

Therefore, the total instantaneous collector current of transistor Q_2 can be obtained by replacing ωt by $(\omega t + \pi)$ in Equation (7.95)

$$i_{c_2} = I_{CQ} + B_0 + B_1 \cos(\omega t + \pi) + B_2 \cos 2(\omega t + \pi) + B_3 \cos 3(\omega t + \pi) + B_4 \cos 4(\omega t + \pi) + \dots \quad (7.97)$$

$$\therefore i_{c_2} = I_{CQ} + B_0 + B_1 \cos(\omega t + \pi) + B_2 \cos(2\omega t + 2\pi) + B_3 \cos(3\omega t + 3\pi) + B_4 \cos(4\omega t + 4\pi) + \dots \quad (7.98)$$

We know that

$$\cos(\theta + n\pi) = \begin{cases} \cos\theta & \text{for even } n \\ -\cos\theta & \text{for odd } n \end{cases} \quad (7.99)$$

Applying Equation (7.99) to Equation (7.98), we have

$$i_{c_2} = I_{CQ} + B_0 - B_1 \cos \omega t + B_2 \cos 2\omega t - B_3 \cos 3\omega t + B_4 \cos 4\omega t + \dots \quad (7.100)$$

From Equation (7.94)

$$i_L = \frac{N_1}{N_2} \{ (I_{CQ} + B_0 + B_1 \cos \omega t + B_2 \cos 2\omega t + B_3 \cos 3\omega t + B_4 \cos 4\omega t + \dots) - (I_{CQ} + B_0 - B_1 \cos \omega t + B_2 \cos 2\omega t - B_3 \cos 3\omega t + B_4 \cos 4\omega t + \dots) \} \quad (7.101)$$

$$\therefore i_L = \frac{N_1}{N_2} (2B_1 \cos \omega t + 2B_3 \cos 3\omega t + \dots) \quad (7.102)$$

$$\text{or} \quad i_L = \frac{2N_1}{N_2} (B_1 \cos \omega t + B_3 \cos 3\omega t + \dots) \quad (7.103)$$

Observe from, Equation (7.103) that the output current is free from even harmonic components. The principal source of distortion is the third harmonic component which is very small in comparison with the fundamental. These results are possible only if the transistors Q_1 and Q_2 are identical. If their characteristics differ, then complete cancellation of even harmonic components will not take place.

Advantages of push-pull configuration :

Following are the advantages of push-pull configuration.

- The output current is free from even harmonic components and therefore the circuit gives more output per transistor for a specified amount of distortion.
- There is no dc current in the primary winding of the output transformer. This eliminates core saturation. Core saturation introduces non linear distortion, which arises from the curvature of the transformer magnetisation curve. Note that core saturation may occur in class A single ended transformer coupled amplifier as indicated in Section 7.2.2.
- V_{CC} is generally obtained from an ac source using rectifier and filter. Inadequate filtering gives rise to ripple voltages in V_{CC} . In the push-pull configuration the ripple currents produced by these ripple voltages are in opposite directions in the primary of the output transformer and therefore cancel out.

7.6.2 Phase Splitting Circuits

In the push pull amplifier circuit of Fig. 7.16, the input transformer was used to provide two signals of equal magnitude and opposite polarity. An alternate circuit which can be used for this purpose is shown in Fig. 7.18.

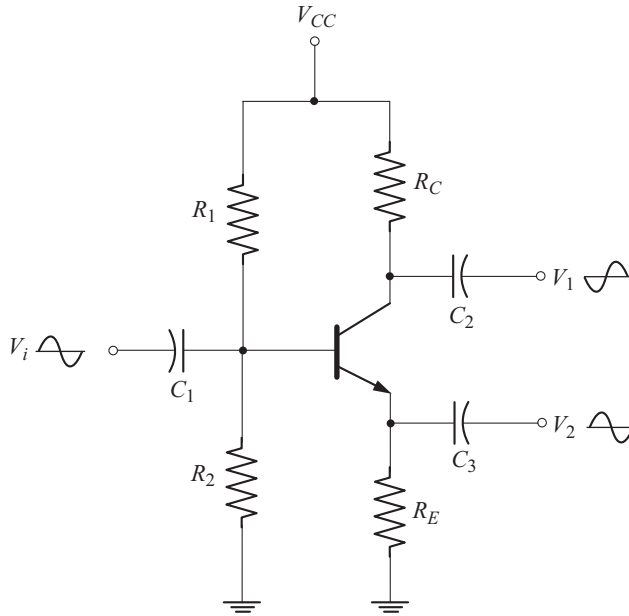


Fig. 7.18 BJT phase splitting circuit

Due to voltage follower action,

$$V_2 \approx V_i$$

Since the emitter resistor is unbypassed, the voltage gain of the CE stage is

$$\frac{V_1}{V_i} \approx -\frac{R_C}{R_E}$$

If R_C is taken equal to R_E , then

$$V_1 \approx -V_i \quad (7.104)$$

Note that V_1 and V_i are equal in magnitude but opposite in phase. Thus the voltages V_1 and V_2 are equal in magnitude with 180° phase shift between them. Another phase splitting circuit using op-amp is shown in Fig. 7.19.

The inverting amplifier introduces 180° phase shift whereas the phase shift introduced by the voltage follower is zero degrees. Thus V_1 and V_2 are 180° out of phase.

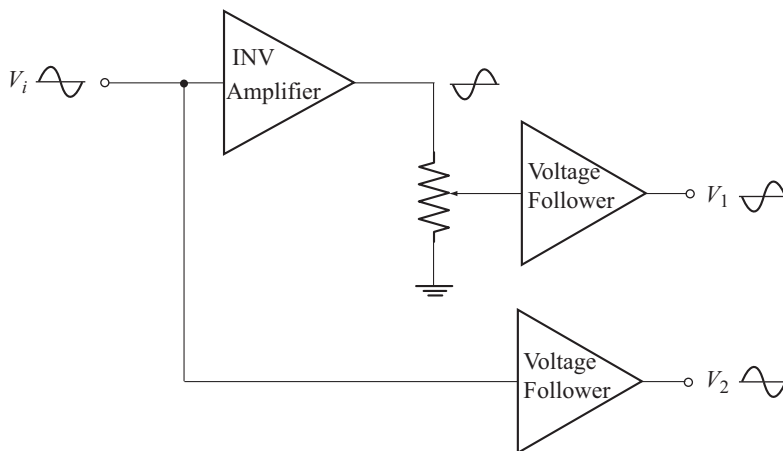


Fig. 7.19 Phase splitting network using op-amp

7.6.3 Conversion Efficiency of Class B Push-pull Power Amplifier

Assuming that the dynamic transfer curve is linear, the output waveforms can be constructed as shown in Fig. 7.20 for a single transistor, say Q_1 .

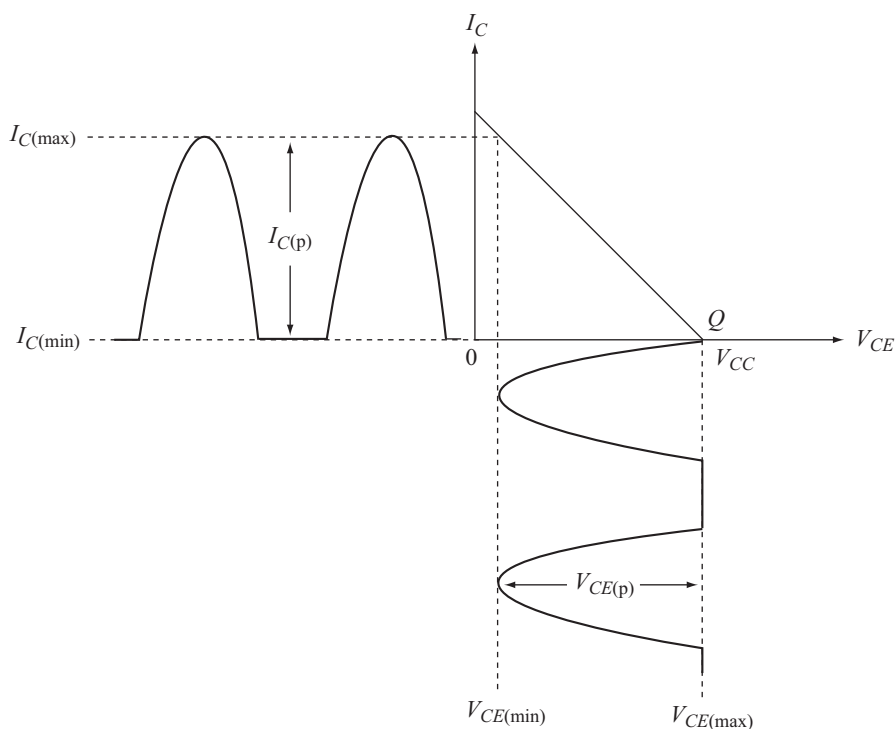


Fig. 7.20 Output voltage and current waveforms in a single transistor of class B push-pull amplifier

The conversion efficiency is given by

$$\% \eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100\% \quad (7.105)$$

AC Output Power [$P_{o(ac)}$]

$$P_{o(ac)} = \frac{V_{CE(p)} I_{C(p)}}{2} \quad (7.106)$$

From Fig. 7.20,

$$\begin{aligned} V_{CE(p)} &= V_{CE(max)} - V_{CE(min)} \\ \text{But } V_{CE(max)} &= V_{CC} \\ \therefore V_{CE(p)} &= V_{CC} - V_{CE(min)} \end{aligned} \quad (7.107)$$

DC input power [$P_{i(dc)}$]

The dc power input to the power amplifier is given by

$$P_{i(dc)} = V_{CC} I_{dc} \quad (7.108)$$

where I_{dc} is the average or the dc current drawn from the power supply V_{CC} . Note that the collector current waveform of each transistor is a half rectified sinusoid with a peak value, $I_{C(p)}$.

Thus the average current in each transistor is $\frac{I_{C(p)}}{\pi}$. Since there are two transistors, the dc current drawn from the supply V_{CC} , by both the transistors is

$$\begin{aligned} \therefore I_{dc} &= 2 [\text{average current in each transistor}] \\ &= \frac{2 I_{C(p)}}{\pi} \end{aligned} \quad (7.109)$$

Using this relation in Equation (7.108) we have

$$P_{i(dc)} = \frac{2}{\pi} V_{CC} I_{C(p)} \quad (7.110)$$

Using Equations (7.106) and (7.110) in Equation (7.105) we get

$$\begin{aligned} \% \eta &= \frac{\left(\frac{V_{CE(p)} I_{C(p)}}{2} \right)}{\frac{2}{\pi} V_{CC} I_{C(p)}} \times 100\% \\ \% \eta &= \frac{\pi}{4} \frac{V_{CE(p)}}{V_{CC}} \times 100\% \end{aligned} \quad (7.111)$$

Efficiency is maximum when

$$V_{CE(p)} = V_{CC} \quad (7.112)$$

This condition occurs when

$$V_{CE(\min)} = 0 \quad [\text{From equation 7.107}]$$

Using Equation (7.112) in Equation (7.111) we have

$$\% \eta_{\max} = \frac{\pi}{4} \times 100\% = 78.54\% \quad (7.113)$$

Observe that the efficiency is higher than that of class A transformer coupled power amplifier which is 50%. This increase in the efficiency results from the fact that in the class B push-pull power amplifier, under no signal condition, there is no power dissipation since $I_{CQ} = 0$ and therefore no current is drawn from V_{CC} .

7.6.4 Power Dissipated by Output Transistors

In class B push-pull power amplifier, the dc power drawn from the source, V_{CC} , is equal to the sum of the ac power developed at the collector and the power dissipated in the collector circuit of the output transistors.

$$P_{i(dc)} = P_{o(ac)} + P_{2Q} \quad (7.114)$$

$$\text{or} \quad P_{2Q} = P_{i(dc)} - P_{o(ac)} \quad (7.115)$$

where P_{2Q} is the power dissipated by the two output power transistors. The power dissipated by each transistor is then

$$P_Q = \frac{P_{2Q}}{2} \quad (7.116)$$

P_{2Q} is also called as the collector dissipation, since the power is dissipated in the collector circuit of the transistors.

7.6.5 Maximum Power Considerations

Now let us calculate the maximum ac output power, maximum dc input power and maximum power dissipation in the output transistors.

Maximum ac Output Power

The ac output power is given by

$$P_{o(ac)} = \frac{V_{CE(p)} I_{C(p)}}{2} \quad (7.117)$$

$$\text{where,} \quad I_{C(p)} = \frac{V_{CE(p)}}{R'_L} \quad (7.118)$$

where, R'_L is the load resistance reflected to primary.

Using Equation (7.118) in (7.117), we get

$$P_{o(ac)} = \frac{V_{CE(p)}^2}{2R'_L} \quad (7.119)$$

The ac output power is maximum when, $V_{CE(p)} = V_{CC}$. Now the maximum ac output power is

$$P_{o(ac)max} = \frac{V_{CC}^2}{2R'_L} \quad (7.120)$$

The corresponding peak ac current is

$$I_{C(p)} = \frac{V_{CC}}{R'_L} \quad (7.121)$$

Maximum dc Power Input

The dc input power is given by

$$P_{i(dc)} = V_{CC} I_{dc} \quad (7.122)$$

$$\text{where } I_{dc} = \frac{2}{\pi} I_{C(p)}$$

$$I_{dc} \text{ is maximum when } I_{C(p)} = \frac{V_{CC}}{R'_L}$$

Now the maximum dc current is

$$I_{dc(max)} = \frac{2}{\pi} \frac{V_{CC}}{R'_L} \quad (7.123)$$

Using this relation in Equation (7.122) we get

$$P_{i(dc)max} = \frac{2}{\pi} \frac{V_{CC}^2}{R'_L} \quad (7.124)$$

Maximum Circuit Efficiency

$$\begin{aligned} \% \eta_{max} &= \frac{P_{o(ac)max}}{P_{i(dc)max}} \times 100\% \\ &= \frac{\frac{V_{CC}^2}{2R'_L}}{\frac{2}{\pi} \frac{V_{CC}^2}{R'_L}} \times 100\% \\ \% \eta_{max} &= \frac{\pi}{4} \times 100\% = 78.54\% \end{aligned} \quad (7.125)$$

Note that to obtain maximum efficiency, the input signal should result in maximum output swing i.e., $V_{CE(p)} = V_{CC}$. Otherwise efficiency will be less than 78.54%.

Maximum Power Dissipation

For class B operation, the maximum power dissipated by the two output transistors occurs when

$$V_{CE(p)} = \frac{2}{\pi} V_{CC} = 0.636 V_{CC} \quad (7.126)$$

The maximum power dissipation is given by

$$P_{2Q(\max)} = \frac{2 V_{CC}^2}{\pi^2 R'_L} \quad (7.127)$$

These results are derived in example (7.27).

It is important to note that

- Maximum ac output power occurs when $V_{CE(p)} = V_{CC}$
- Maximum power dissipation in the output transistors occurs when $V_{CE(p)} = 0.636 V_{CC}$

Thus maximum power dissipation in the output transistors does not occur when the ac output power is maximum but it occurs at an ac output power which is less than its maximum value.

Example 7.27

Show that in class B push pull power amplifier:

- (a) The maximum power dissipated by the two output transistors occurs when

$$V_{CE(p)} = 0.636 V_{CC}$$

- (b) The maximum transistor power dissipation is,

$$P_{2Q(\max)} = \frac{2 V_{CC}^2}{\pi^2 R'_L}$$

- (c) $P_{2Q(\max)} = 0.4 P_{o(ac) \max}$

Solution

(a) In class B push-pull power amplifier, the dc power drawn from the source V_{CC} is the sum of the ac power developed at the collector and the power dissipated in the collector circuit.

$$P_{i(dc)} = P_{o(ac)} + P_{2Q} \quad (A)$$

where P_{2Q} is the power dissipated in the collector circuit

$$\therefore P_{2Q} = P_{i(dc)} - P_{o(ac)} \quad (B)$$

From Equation (7.110)

$$P_{i(dc)} = \frac{2}{\pi} V_{CC} I_{C(p)}$$

But
$$I_{C(p)} = \frac{V_{CE(p)}}{R'_L}$$

$$\therefore P_{i(dc)} = \frac{2 V_{CC} V_{CE(p)}}{\pi R'_L} \quad (C)$$

Also
$$P_{o(ac)} = \frac{V_{CE(p)} I_{C(p)}}{2}$$

$$\text{or} \quad P_{o(ac)} = \frac{V_{CE(p)}^2}{2 R'_L} \quad (D)$$

Substituting equations (C) and (D) in Equation (B).

$$P_{2Q} = \frac{2 V_{CC} V_{CE(p)}}{\pi R'_L} - \frac{V_{CE(p)}^2}{2 R'_L} \quad (E)$$

Collector dissipation is zero under the following two conditions

1. When $V_{CE(p)} = 0$

This happens under zero signal condition

2. When $\frac{2 V_{CC} V_{CE(p)}}{\pi R'_L} = \frac{V_{CE(p)}^2}{2 R'_L}$

$$\Rightarrow V_{CE(p)} = \frac{4}{\pi} V_{CC}$$

Thus, the collector dissipation is zero at $V_{CE(p)} = 0$ and $V_{CE(p)} = \frac{4}{\pi} V_{CC}$. The maximum collector dissipation occurs at a value of $V_{CE(p)}$ in the range $0 < V_{CE(p)} < \frac{4}{\pi} V_{CC}$

Let us now find the value of $V_{CE(p)}$ corresponding to maximum power dissipation by setting

$$\frac{dP_{2Q}}{dV_{CE(p)}} = 0$$

From Equation (E)

$$\begin{aligned} \frac{dP_{2Q}}{dV_{CE(p)}} &= \frac{2 V_{CC}}{\pi R'_L} - \frac{V_{CE(p)}}{R'_L} = 0 \\ \therefore V_{CE(p)} &= \frac{2 V_{CC}}{\pi} = 0.636 V_{CC} \end{aligned} \quad (F)$$

(b) The maximum collector dissipation, $P_{2Q \max}$ is obtained by substituting Equation (F) in Equation (E)

$$\begin{aligned} P_{2Q \max} &= \frac{2 V_{CC}}{\pi R'_L} \left(\frac{2 V_{CC}}{\pi} \right) - \left(\frac{1}{2 R'_L} \right) \left(\frac{2 V_{CC}}{\pi} \right)^2 \\ &= \frac{4 V_{CC}^2}{\pi^2 R'_L} - \frac{2 V_{CC}^2}{\pi^2 R'_L} \\ \therefore P_{2Q \max} &= \frac{2 V_{CC}^2}{\pi^2 R'_L} \end{aligned} \quad (G)$$

Using Equation (E), the plot of P_{2Q} versus $V_{CE(p)}$ is shown in Fig. A.

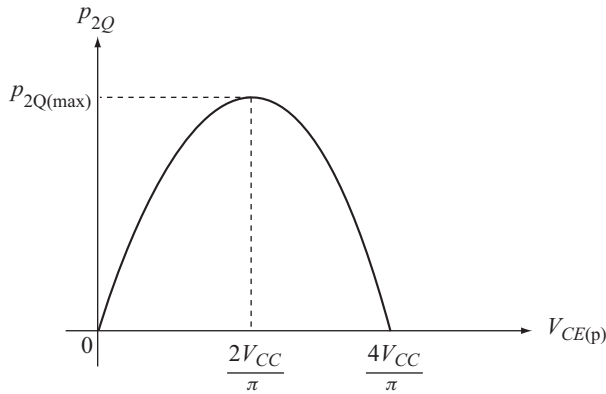


Fig. A Plot of P_{2Q} versus $V_{CE(p)}$ for class B push-pull power amplifier

From Fig. A, we observe that

- (i) At zero signal ($V_{CE(p)} = 0$) collector dissipation is zero
- (ii) The collector dissipation increases and reaches a maximum when $V_{CE(p)} = \frac{2V_{CC}}{\pi}$
- (iii) The collector dissipation then decreases and becomes zero once again when $V_{CE(p)} = \frac{4V_{CC}}{\pi}$

(c) From Equation (7.120)

$$P_{o(ac)max} = \frac{V_{CC}^2}{2R'_L} \quad (H)$$

From equations (G) and (H)

$$\begin{aligned} \frac{P_{2Q(max)}}{P_{o(ac)max}} &= \frac{\left(\frac{2V_{CC}^2}{\pi^2 R'_L} \right)}{\left(\frac{V_{CC}^2}{2R'_L} \right)} = \frac{4}{\pi^2} \\ \therefore \frac{P_{2Q(max)}}{P_{o(ac)max}} &\approx 0.4 \\ \text{or } P_{2Q(max)} &= 0.4 P_{o(ac)max} \quad (I) \end{aligned}$$

We are now going to make a very interesting observation. Suppose it is required to deliver a maximum of 20 W to the load from a class B push-pull power amplifier. From Equation (I), the maximum collector dissipation is 8 W, which is equally shared by Q_1 and Q_2 . Thus, the collector dissipation in each transistor is 4 W, which is 1/5 of the total ac power delivered to the load.

Thus in class B push-pull power amplifier, the ac power delivered to the load is five times the collector dissipation in each transistor.

7.6.6 Class B Push-pull Power Amplifier using Complementary Symmetry Transistor Pair

Figure 7.21 shows the circuit of class B push-pull power amplifier using complementary symmetry transistor pair. Q_1 is an npn transistor and Q_2 is a pnp transistor. The transistors Q_1 and Q_2 are assumed to have identical characteristics. Note that both the transistors are biased at cut-off, since no dc bias is applied to the base-emitter circuit.

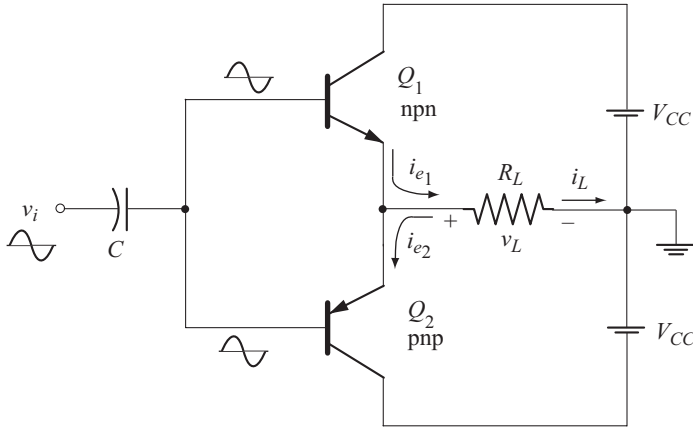


Fig. 7.21 Complementary symmetry push-pull circuit

During the positive half cycle of the input signal v_i , Q_1 will be biased in to conduction and Q_2 remains off.

$$\therefore i_L = i_{e1}$$

The negative half cycle of v_i , biases Q_2 into conduction and Q_1 remains off.

$$\therefore i_L = -i_{e2}$$

Combining these,

$$i_L = \begin{cases} i_{e1} & \text{for } 0 \leq \omega t \leq \pi \\ -i_{e2} & \text{for } \pi \leq \omega t \leq 2\pi \end{cases} \quad (7.128)$$

It is important to note that, both Q_1 and Q_2 are operating as emitter followers, with a voltage gain nearing unity. As a result the load voltage is essentially the same as the input voltage.

The analysis given in sections 7.6.3 to 7.6.5 for transformer coupled class B push-pull power amplifier can be directly applied to this circuit. Since the circuit operates without output transformer we have to apply the following modifications.

$$\begin{aligned}
 V_{CE(p)} &= V_{L(p)} \\
 I_{C(p)} &= I_{L(p)} \\
 \text{and} \quad R'_L &= R_L
 \end{aligned}$$

The output voltage and current waveforms are same as that given in Fig. 7.17 for transformer coupled class B push-pull power amplifier, with i_{c1} and i_{c2} replaced by i_{e1} and i_{e2} respectively.

The advantages of this circuit are:

Due to the absence of input and output transformers the circuit is

- Simple
- Less expensive
- Less bulky and
- light weight.

The main drawbacks of this circuit are:

- The circuit requires two dc power supplies.
- Since both transistors are biased at cut-off cross over distortion is present.
- It is very difficult to get matched transistor pair.
- Poor impedance matching due to the absence of output transformer.
- Even harmonic distortion may be expected if the two transistors are not perfectly matched.
- The ripple content of the dc supply, which is present due to inadequate filtering, will reach the load.

7.6.7 Practical Complementary-Symmetry Class B Push-pull Circuit using Darlington Transistors

Figure 7.22 shows the circuit of practical complementary-symmetry class B push-pull circuit using darlington transistors. During the positive half cycle of v_i , the npn Darlington transistor conducts and the *pnp* Darlington transistor conducts for the negative half cycle of v_i . The waveforms of Fig. 7.17 applies to this circuit also.

The main features of this circuit are:

- The biasing resistors R_1 and R_2 keep the Darlington transistors in the verge of conduction. As a result, cross over distortion is absent.
- The low output impedance of the Darlington transistors properly match the low impedance of the load which is usually a loud speaker.
- The Darlington transistors provide higher output current.
- A small amount of negative feedback provided through the emitter resistors R_E (typically $1\ \Omega$) helps to keep the harmonic distortion at a minimum.

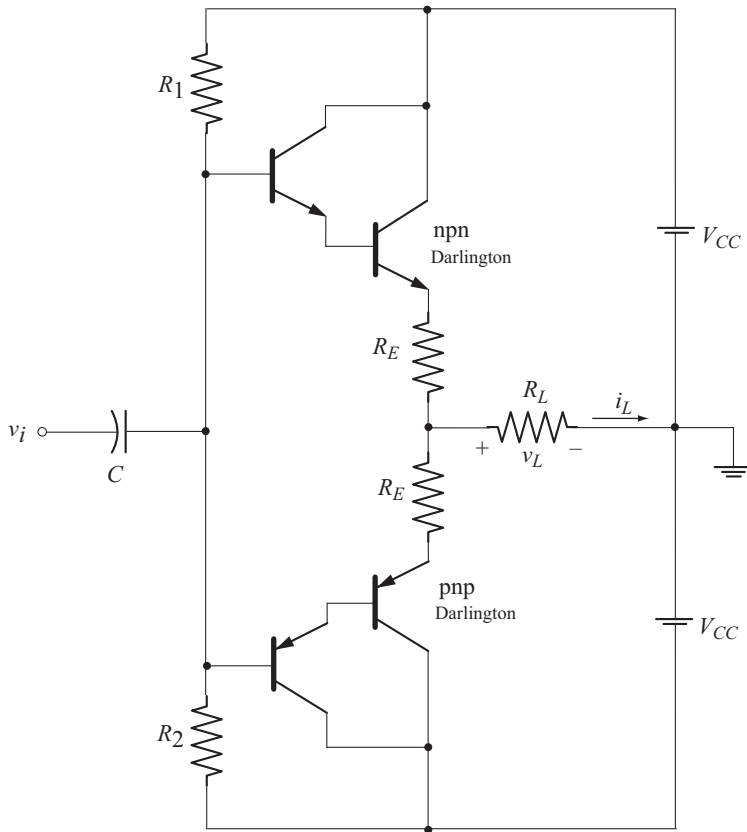


Fig. 7.22 Circuit of practical complementary-symmetry class B push-pull circuit

7.6.8 Quasi-Complementary Class B Push-pull Power Amplifier

A practical power amplifier is required to deliver large current in to low impedance load. The current driving capacity of a pnp transistor is less than that of npn transistor. Hence it is preferable to use npn transistors for both high-current-output devices.

Figure 7.23 shows the circuit of quasi-complementary push-pull class B power amplifier. The push pull operation is provided by the complementary transistors Q_1 and Q_2 . Q_3 and Q_4 are the matched output transistors.

During the positive half cycle of the input signal v_i , Q_1 is driven into conduction while Q_2 remains off. The load current is supplied by the Darlington transistor consisting of Q_1 and Q_3 .

During the negative half cycle of v_i , Q_2 is driven into conduction while Q_1 remains off. Now the feedback pair comprising of Q_2 and Q_4 supplies the load current.

Biasing resistors R_1 , R_2 and R_3 keeps the Darlington pair and the feedback pair in the verge of conduction. Resistor R_2 can be adjusted to minimize cross over distortion.

Both the Darlington pair and the feedback pair operates with low output impedance. This ensures a proper impedance matching between the amplifier output and the load. The quasi-complementary push-pull amplifier is the most popular form of power amplifier. The waveforms given in Fig. 7.17 are also applicable to this circuit.

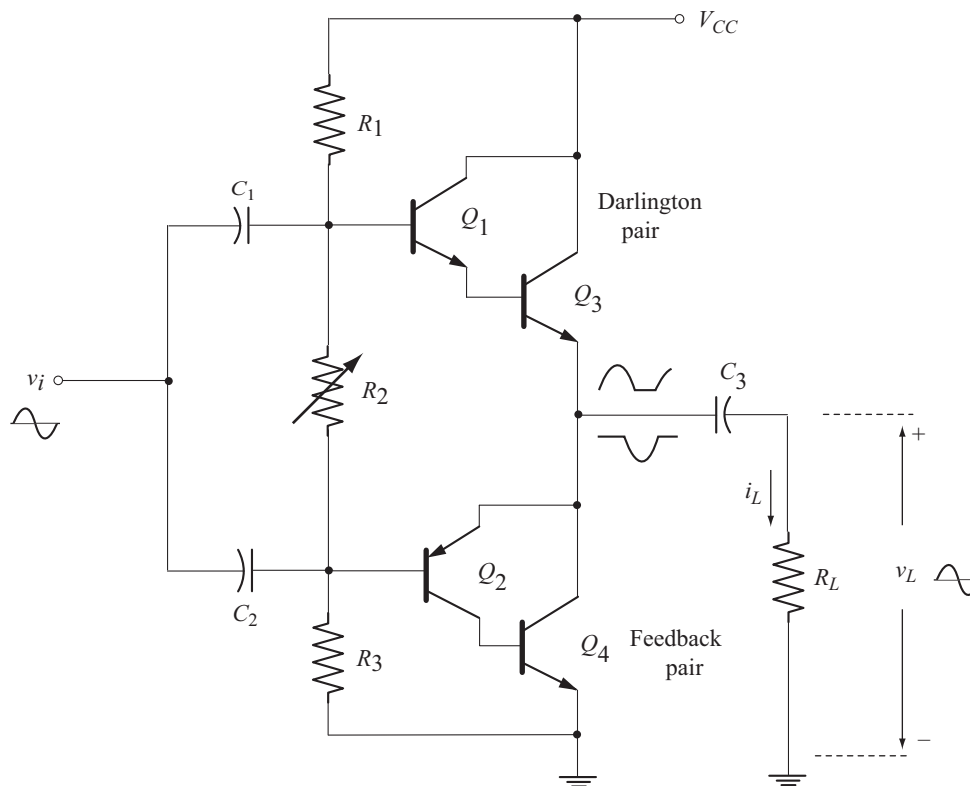


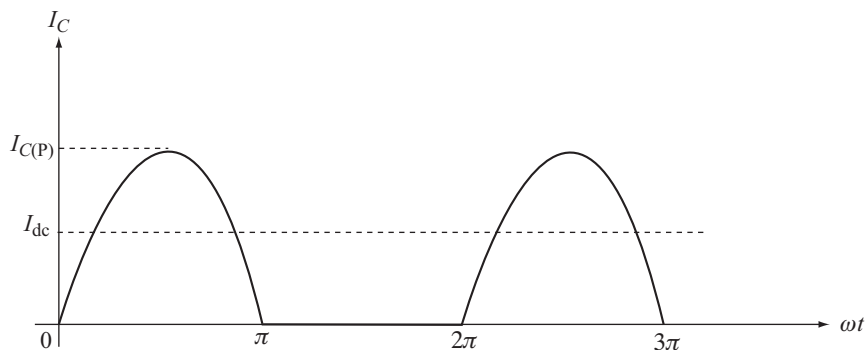
Fig. 7.23 Quasi-complementary class B push-pull power amplifier

Example 7.28

A single transistor is operating as an ideal class B amplifier with $1\text{ k}\Omega$ load. A dc meter in the circuit reads 10 mA . How much signal power is delivered to the load.

Solution

Since a single transistor is being used, the collector current waveform is shown below.



$$\begin{aligned}\text{Given} \quad I_{dc} &= 10 \text{ mA} \\ R_L &= 1 \text{ k}\Omega\end{aligned}$$

Since I_C is a half rectified sine wave,

$$\begin{aligned}I_{dc} &= \frac{I_{C(p)}}{\pi} \\ \therefore I_{C(p)} &= \pi I_{dc} = 31.4 \text{ mA}\end{aligned}$$

Power delivered to the load,

$$P_{o(ac)} = I_{C(rms)}^2 R'_L$$

Since the output transformer is not specified, assume

$$R'_L = R_L$$

For half rectified sine wave,

$$\begin{aligned}I_{C(rms)} &= \frac{I_{C(p)}}{2} \\ \therefore P_{o(ac)} &= \frac{I_{C(p)}^2}{4} \times R_L = \frac{(31.4 \text{ mA})^2 (1 \text{ k}\Omega)}{4} \\ \text{or} \quad P_{o(ac)} &= 0.246 \text{ W}.\end{aligned}$$

Example 7.29

An ideal class B push–pull power amplifier with input and output transformers, has $V_{CC} = 20 \text{ V}$, $N_2 = 2 N_1$ and $R_L = 20 \text{ }\Omega$. The transistors have $h_{FE} = 20$. Let the input be sinusoidal. For the maximum output signal at $V_{CE(p)} = V_{CC}$, determine:

- the output signal power
- the collector dissipation in each transistor
- conversion efficiency

Solution

Refer Fig. 7.16

$$\begin{aligned}\text{Given} \quad V_{CC} &= 20 \text{ V} \\ R_L &= 20 \text{ }\Omega \\ N_2 &= 2N_1 \\ \text{or} \quad \frac{N_1}{N_2} &= 0.5 \\ V_{CE(p)} &= V_{CC} = 20 \text{ V}\end{aligned}$$

$$(a) \quad P_{o(ac)} = \frac{V_{CE(p)}^2}{2 R'_L} \quad (A)$$

$$R'_L = \left(\frac{N_1}{N_2} \right)^2 R_L = 0.5^2 \times 20 \, \Omega = 5 \, \Omega$$

From Equation (A)

$$(b) \quad P_{o(ac)} = \frac{(20 \text{ V})^2}{(2)(5 \, \Omega)} = 40 \text{ W}$$

$$P_{2Q} = P_{i(dc)} - P_{o(ac)} \quad (B)$$

$$P_{i(dc)} = \frac{2}{\pi} V_{CC} I_{C(P)}$$

$$\text{Using} \quad I_{C(P)} = \frac{V_{CE(p)}}{R'_L} = \frac{V_{CC}}{R'_L}$$

$$\text{We get,} \quad P_{i(dc)} = \frac{2}{\pi} \frac{V_{CC}^2}{R'_L} = \frac{2}{\pi} \frac{(20 \text{ V})^2}{5 \, \Omega} = 50.93 \text{ W}$$

From Equation (B),

$$P_{2Q} = 50.93 \text{ W} - 40 \text{ W} = 10.93 \text{ W}$$

$$P_Q = \frac{P_{2Q}}{2} = \frac{10.93 \text{ W}}{2} = 5.46 \text{ W}$$

$$(c) \quad \% \eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100\% = \frac{40 \text{ W}}{50.93 \text{ W}} \times 100\%$$

$$\% \eta = 78.54\% \text{ as expected}$$

Example 7.30

A class B push-pull amplifier operating with $V_{CC} = 25 \text{ V}$ provides a 22 V peak signal to an $8 \, \Omega$ load. Find

- rms and peak load (output) currents.
- rms and peak collector currents.
- dc current drawn from the supply
- Input power
- Output power
- Circuit efficiency
- Power dissipation in the transistors

Solution

Given

$$V_{CC} = 25 \text{ V}$$

$$V_{L(p)} = 22 \text{ V (peak load voltage)}$$

$$R_L = 8\Omega$$

Note that the output transformer is not specified. For transformer less class B power amplifier

$$\begin{aligned} V_{CE(p)} &= V_{L(p)} \\ I_{C(p)} &= I_{L(p)} \\ R'_L &= R_L \end{aligned}$$

(a) peak load current is

$$I_{L(p)} = \frac{V_{L(p)}}{R_L} = \frac{22\text{ V}}{8\Omega} = 2.75\text{ A}$$

rms load current is

$$I_{L(\text{rms})} = \frac{I_{L(p)}}{\sqrt{2}} = \frac{2.75\text{ A}}{\sqrt{2}} = 1.945\text{ A}$$

(b) rms and peak collector currents are same as rms and peak load currents

$$\begin{aligned} \therefore I_{C(\text{rms})} &= I_{L(\text{rms})} = 1.945\text{ A} \\ I_{C(p)} &= I_{L(p)} = 2.75\text{ A} \end{aligned}$$

(c) dc current drawn from the supply is

$$I_{dc} = \frac{2}{\pi} I_{C(p)} = \frac{2}{\pi} (2.75\text{ A}) = 1.75\text{ A}$$

$$\text{dc current drawn by each transistor} = \frac{I_{dc}}{2} = 0.875\text{ A}$$

(d) Input power refers to the dc input power.

$$P_{i(dc)} = V_{CC} I_{dc} = (25\text{ V})(1.75\text{ A}) = 43.75\text{ W}$$

(e) Output power refers to ac output power

$$P_{o(ac)} = \frac{V_{CE(p)} I_{C(p)}}{2} = \frac{(22\text{ V})(2.75\text{ A})}{2} = 30.25\text{ W}$$

(f) Circuit efficiency refers to conversion efficiency

$$\% \eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100\% = \frac{30.25\text{ W}}{43.75\text{ W}} \times 100\% = 69.143\%.$$

(g) Power dissipation in the output transistors is

$$P_{2Q} = P_{i(dc)} - P_{o(ac)} = 43.75\text{ W} - 30.25\text{ W} = 13.5\text{ W}$$

Power dissipation in each transistor is

$$P_Q = \frac{P_{2Q}}{2} = \frac{13.5\text{ W}}{2} = 6.75\text{ W}$$

Example 7.31

For a class B push-pull power amplifier with $V_{CC} = 25$ V driving an $8\ \Omega$ load, find

- Maximum input power
- Maximum output power
- Maximum circuit efficiency
- Maximum collector dissipation and
- The input voltage at which maximum power dissipation occurs.

Solution

Given

$$V_{CC} = 25\text{ V}$$

$$R_L = 8\ \Omega$$

Since output transformer is not specified, assume

$$R'_L = R_L = 8\ \Omega$$

- (a) From Equation (7.124), the maximum input power is

$$P_{i(dc)max} = \frac{2}{\pi} \frac{V_{CC}^2}{R'_L} = \frac{2}{\pi} \frac{(25\text{ V})^2}{8\ \Omega} = 49.74\text{ W}$$

- (b) From Equation (7.120) the maximum output power is

$$P_{o(ac)max} = \frac{V_{CC}^2}{2 R'_L} = \frac{(25\text{ V})^2}{(2)(8\ \Omega)} = 39.06\text{ W}$$

- (c) Maximum circuit efficiency is

$$\begin{aligned} \% \eta_{max} &= \frac{P_{o(ac)max}}{P_{i(dc)max}} \times 100\% \\ &= \frac{39.06\text{ W}}{49.74\text{ W}} \times 100\% = 78.53\% \text{ as expected} \end{aligned}$$

- (d) From Equation (7.127), the maximum collector dissipation is

$$P_{2Q(max)} = \frac{2}{\pi^2} \frac{V_{CC}^2}{R'_L} = \frac{2}{\pi^2} \frac{(25\text{ V})^2}{8\ \Omega} = 15.83\text{ W}$$

Maximum collector dissipation in each transistor is

$$P_{Q(max)} = \frac{P_{2Q(max)}}{2} = \frac{15.83\text{ W}}{2} = 7.915\text{ W}$$

- (e) From Equation (7.126), the input voltage at which maximum power dissipation occurs is

$$\begin{aligned} V_{i(p)} &= V_{CE(p)} = (0.636) V_{CC} \\ &= (0.636) (25\text{ V}) = 15.9\text{ V} \end{aligned}$$

Example 7.32

Calculate the efficiency of class B push-pull power amplifier for a supply voltage of $V_{CC} = 22 \text{ V}$, driving a 4Ω load with peak output voltages of

- (a) $V_{L(p)} = 22 \text{ V}$
- (b) $V_{L(p)} = 20 \text{ V}$
- (c) $V_{L(p)} = 4 \text{ V}$
- (d) Compare and comment on the results.

Solution

Given $V_{CC} = 22 \text{ V}$

Since output transformer is not specified

$$R'_L = R_L = 4 \Omega$$

From Equation (7.111),

$$\% \eta = \frac{\pi}{4} \frac{V_{CE(p)}}{V_{CC}} \times 100\%$$

For transformerless circuit,

$$V_{CE(p)} = V_{L(p)}$$

$$\therefore \% \eta = \frac{\pi}{4} \frac{V_{L(p)}}{V_{CC}} \times 100\% \quad (\text{A})$$

(a) $V_{L(p)} = 22 \text{ V}$

$$\% \eta = \frac{\pi}{4} \left[\frac{22 \text{ V}}{22 \text{ V}} \right] 100\% = 78.54\%$$

(b) $V_{L(p)} = 20 \text{ V}$

$$\% \eta = \frac{\pi}{4} \left[\frac{20 \text{ V}}{22 \text{ V}} \right] \times 100\% = 71.4\%$$

(c) $V_{L(p)} = 4 \text{ V}$

$$\% \eta = \frac{\pi}{4} \left[\frac{4 \text{ V}}{22 \text{ V}} \right] \times 100\% = 14.27\%$$

(d) Observe from Equation (A) that, efficiency increases with $V_{L(p)}$ and becomes maximum i.e., 78.54 % when $V_{L(p)} = V_{CC}$

Example 7.33

In a class B amplifier, $V_{CE(\min)} = 1 \text{ V}$ and $V_{CC} = 18 \text{ V}$. Calculate the collector circuit efficiency.

Solution

Given $V_{CC} = 18 \text{ V}$ $V_{CE(\min)} = 1 \text{ V}$

$$\% \eta = \frac{\pi}{4} \frac{V_{CE(p)}}{V_{CC}} \times 100 \%$$

From Equation (7.107),

$$\begin{aligned} V_{CE(p)} &= V_{CC} - V_{CE(\min)} \\ &= 18 \text{ V} - 1 \text{ V} = 17 \text{ V} \end{aligned}$$

$$\text{Now } \% \eta = \frac{\pi}{4} \frac{17 \text{ V}}{18 \text{ V}} \times 100\% = 74.18 \%$$

Example 7.34

Calculate the power dissipated in the individual transistors of a class B push-pull power amplifier if $V_{CC} = 20 \text{ V}$ and $R_L = 4 \Omega$.

Solution

$$\text{Given } V_{CC} = 20 \text{ V}$$

Since output transformer is not specified,

$$R'_L = R_L = 4 \Omega$$

Collector dissipation

$$P_{2Q} = P_{i(dc)} - P_{o(ac)} \quad (\text{A})$$

Since the output signal amplitude is not given, let us assume maximum power output

$$\begin{aligned} \therefore V_{CE(p)} &= V_{CC} \\ P_{o(ac)} &= \frac{V_{CE(p)}^2}{2R'_L} = \frac{V_{CC}^2}{2R_L} = \frac{(20 \text{ V})^2}{(2)(4 \Omega)} = 50 \text{ W} \end{aligned}$$

$$P_{i(dc)} = \frac{2}{\pi} V_{CC} I_{C(p)}$$

$$\text{But } I_{C(p)} = \frac{V_{CE(p)}}{R'_L} = \frac{V_{CC}}{R_L}$$

$$\therefore P_{i(dc)} = \frac{2 V_{CC}^2}{\pi R_L} = \frac{2 \times (20 \text{ V})^2}{\pi \times 4 \Omega} = 63.66 \text{ W}$$

Substituting in Equation (A)

$$P_{2Q} = 63.66 \text{ W} - 50 \text{ W} = 13.66 \text{ W}$$

Power dissipation in each transistor is

$$P_Q = \frac{P_{2Q}}{2} = \frac{13.66 \text{ W}}{2} = 6.83 \text{ W}$$

Example 7.35

Calculate the peak power dissipated in each transistor of a class B push pull power amplifier if $V_{CC} = 15 \text{ V}$ and $R'_L = 5 \Omega$

Solution

Given

$$V_{CC} = 15 \text{ V} \quad R'_L = 5 \Omega$$

Peak power dissipation in the output transistors is

$$\begin{aligned} P_{2Q(\max)} &= \frac{2}{\pi^2} \frac{V_{CC}^2}{R'_L} \\ &= \frac{2}{\pi^2} \frac{(15 \text{ V})^2}{5 \Omega} = 9.12 \text{ W} \end{aligned}$$

Peak power dissipation in each transistor is

$$\begin{aligned} P_{Q(\max)} &= \frac{P_{2Q(\max)}}{2} \\ &= \frac{9.12 \text{ W}}{2} = 4.56 \text{ W} \end{aligned}$$

Example 7.36

A class B transformer-coupled push pull power amplifier is to supply 5W to a 18Ω load with $V_{CC} = 30 \text{ V}$. Assume transformer efficiency of 75%. Determine:

- Turns ratio of the output transformer
- Power dissipation in each transistor and
- Conversion efficiency

Solution

Given

$$V_{CC} = 30 \text{ V}$$

Power delivered to the load (secondary),

$$\begin{aligned} P_L &= 5 \text{ W} \\ R_L &= 18 \Omega \\ \eta_{\text{TFR}} &= 0.75 \\ \eta_{\text{TFR}} &= \frac{\text{Power delivered to secondary}}{\text{Power developed in primary}} \\ &= \frac{P_L}{P_{o(\text{ac})}} \\ \therefore P_{o(\text{ac})} &= \frac{P_L}{\eta_{\text{TFR}}} = \frac{5 \text{ W}}{0.75} = 6.67 \text{ W} \end{aligned}$$

$$(a) \quad R'_L = \left[\frac{N_1}{N_2} \right]^2 R_L$$

$$\therefore \quad \frac{N_1}{N_2} = \sqrt{\frac{R'_L}{R_L}} \quad (A)$$

$$\text{But } P_{o(ac)} = \frac{V_{CE(p)}^2}{2 R'_L} \quad (B)$$

$$V_{CE(p)} = V_{CC} - V_{CE(\min)}$$

Taking $V_{CE(\min)} = 0$, for maximum power output,

$$V_{CE(p)} = V_{CC} = 30 \text{ V}$$

$$\text{From (B)} \quad R'_L = \frac{V_{CE(p)}^2}{2 P_{o(ac)}} = \frac{V_{CC}^2}{2 P_{o(ac)}} = \frac{(30 \text{ V})^2}{2 \times 6.67 \text{ W}} = 67.47 \Omega$$

Substituting in (A)

$$\frac{N_1}{N_2} = \sqrt{\frac{67.47 \Omega}{18 \Omega}} = 1.94$$

$$(b) \quad P_{2Q} = P_{i(dc)} - P_{o(ac)} \quad (C)$$

$$P_{i(dc)} = \frac{2}{\pi} V_{CC} I_{C(p)}$$

$$\text{where } I_{C(p)} = \frac{V_{CE(p)}}{R'_L} = \frac{V_{CC}}{R'_L}$$

$$\therefore \quad P_{i(dc)} = \frac{2 V_{CC}^2}{\pi R'_L} = \frac{2 \times (30 \text{ V})^2}{\pi \times 67.47 \Omega} = 8.5 \text{ W}$$

Substituting in (C)

$$P_{2Q} = 8.5 \text{ W} - 6.67 \text{ W} = 1.83 \text{ W}$$

$$\therefore \quad P_Q = \frac{P_{2Q}}{2} = \frac{1.83 \text{ W}}{2} = 0.92 \text{ W}$$

(c) Conversion efficiency

$$\% \eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100 \%$$

$$\% \eta = \frac{6.67}{8.5} \times 100 \% = 78.47 \%$$

as expected for maximum power output condition.

Example 7.37

Analyse the operation of the push pull amplifier shown in Fig. (a).

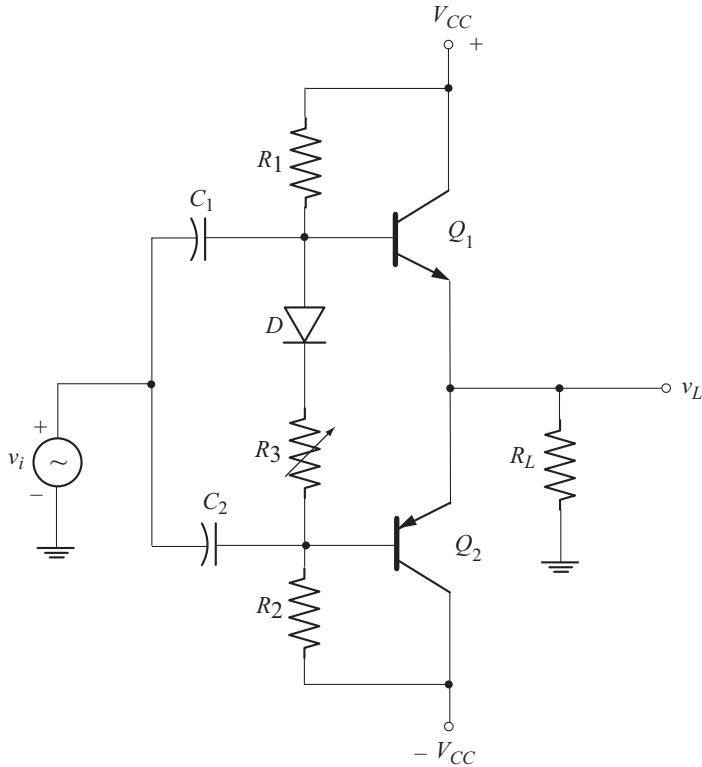


Fig. a

Solution

With R_1 , R_2 , R_3 and D removed, the circuit becomes class B push pull power amplifier without input and output transformer as shown in Fig. (b). Note that this circuit is complementary push-pull class B power amplifier shown in Fig. 7.21 of section 7.6.6.

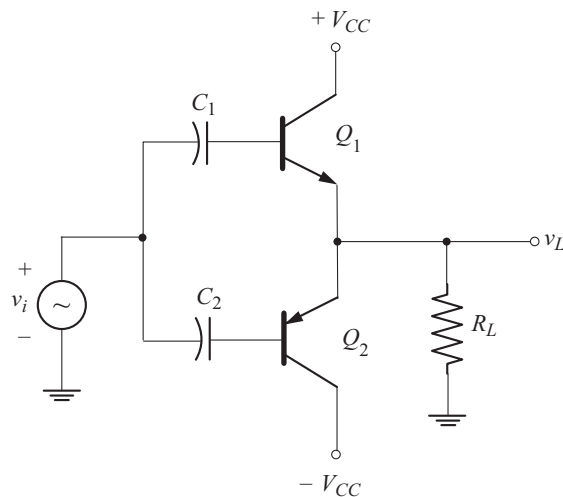


Fig. b

In the circuit of Fig. (b), since $V_{BE} = 0$ for both Q_1 and Q_2 , they are biased at cut-off, which results in cross-over distortion as described in Section 7.1.3. To eliminate cross-over distortion, the transistors should be biased slightly above cut-off to an extent of the base-emitter cut in voltage, typically 0.7 V for silicon transistors.

In the given circuit of Fig. (a), the drop across diode D provides the required forward bias for the base-emitter junction of both the transistors. The diode D_2 is forward biased though resistors R_1 and R_2 connected between V_{CC} and $-V_{CC}$. R_3 is adjusted to minimize cross over distortion.

For ac operation, with reference to Fig. (a), the dc voltages $+V_{CC}$ and $-V_{CC}$ are grounded and the capacitors C_1 and C_2 are shorted. Therefore the collectors of the two transistors are at ground potential. For each transistor, the input is applied between base and ground (collector is at ground) and the output is taken between emitter and ground. Therefore, Q_1 and Q_2 act as emitter-followers. As a result, $v_o = v_i$.

Example 7.38

In the circuit of Example 7.36 Let $\pm V_{CC} = \pm 30$ V, $R_L = 8 \Omega$, $R_1 = R_2 = R_3 = 100 \Omega$, $C_1 = C_2 = 100 \mu\text{F}$. If the rms input voltage is 8 V, calculate

- rms and peak load (output) voltages
- rms and peak load (output) currents
- dc current drawn from the supply
- dc input power
- ac output power
- conversion efficiency and
- power dissipated in each transistor

Solution

Given

$$\pm V_{CC} = \pm 30 \text{ V} \Rightarrow V_{CC} = 30 \text{ V}$$

$$R'_L = R_L = 8 \Omega \quad [\text{No output transformer}]$$

$$\text{rms input voltage} \quad V_{i(\text{rms})} = 8 \text{ V}$$

(a) Since Q_1 and Q_2 are configured as emitter followers, as explained in Example 7.37

$$\text{rms load voltage} = \text{rms input voltage.}$$

$$\text{ie.} \quad V_{L(\text{rms})} = V_{i(\text{rms})} = 8 \text{ V}$$

Peak load voltage is

$$V_{L(p)} = \sqrt{2} V_{L(\text{rms})} = (\sqrt{2})(8 \text{ V}) = 11.31 \text{ V}$$

(b) rms load current is

$$I_{L(\text{rms})} = \frac{V_{L(\text{rms})}}{R_L} = \frac{8 \text{ V}}{8 \Omega} = 1 \text{ A}$$

Peak load current is

$$I_{L(p)} = \frac{V_{L(p)}}{R_L} = \frac{11.31 \text{ V}}{8 \Omega} = 1.414 \text{ A}$$

(c) dc current drawn from the supply is

$$I_{dc} = \frac{2}{\pi} I_{C(p)}$$

For transformer less circuit

$$I_{C(p)} = I_{L(p)} \text{ and}$$

$$V_{CE(p)} = V_{L(p)}$$

$$\therefore I_{dc} = \frac{2}{\pi} I_{L(p)} = \frac{2}{\pi} (1.414 \text{ A}) = 0.9 \text{ A}$$

dc current through each transistor is

$$\frac{I_{dc}}{2} = \frac{0.9 \text{ A}}{2} = 0.45 \text{ A}$$

(d) dc input power is

$$P_{i(dc)} = V_{CC} I_{dc} = (30 \text{ V})(0.9 \text{ A}) = 27 \text{ W}$$

(e) ac output power is

$$\begin{aligned} P_{o(ac)} &= \frac{V_{CE(p)} I_{C(p)}}{2} = \frac{V_{L(p)} I_{L(p)}}{2} \\ &= \frac{(11.31 \text{ V})(1.414 \text{ A})}{2} = 8 \text{ W} \end{aligned}$$

Alternatively,

$$P_{o(ac)} = V_{L(rms)} I_{L(rms)} = (8 \text{ V})(1 \text{ A}) = 8 \text{ W}$$

(f) Conversion efficiency is

$$\% \eta = \frac{P_{o(ac)}}{P_{i(dc)}} \times 100\% = \frac{8 \text{ W}}{27 \text{ W}} \times 100\% = 29.63\%$$

(g) power dissipation in both transistors is

$$P_{2Q} = P_{i(dc)} - P_{o(ac)} = 27 \text{ W} - 8 \text{ W} = 19 \text{ W}$$

Power dissipation in each transistor is

$$P_Q = \frac{P_{2Q}}{2} = \frac{19 \text{ W}}{2} = 9.5 \text{ W}$$

Example 7.39

In the circuit of Example 7.37 let $V_{CC} = 30 \text{ V}$, $R_L = 8 \Omega$. Find

- Maximum dc input power
- Maximum ac output power
- Maximum conversion efficiency
- Input voltage for maximum power operation
- Power dissipated by the output transistors at this voltage

Solution

Given

$$V_{CC} = 30 \text{ V}$$

$$R'_L = R_L = 8 \Omega \quad [\text{No output transformer}].$$

(a) From Equation (7.124), the maximum dc input power is

$$P_{i(dc)\max} = \frac{2}{\pi} \frac{V_{CC}^2}{R_L} = \frac{2}{\pi} \frac{(30 \text{ V})^2}{8 \Omega} = 71.62 \text{ W}$$

(b) From Equation (7.120), the maximum ac output power is

$$P_{o(ac)\max} = \frac{V_{CC}^2}{2R_L} = \frac{(30 \text{ V})^2}{(2)(8 \Omega)} = 56.25 \text{ W}$$

(c) Maximum conversion efficiency is

$$\begin{aligned} \% \eta_{\max} &= \frac{P_{o(ac)\max}}{P_{i(dc)\max}} \times 100\% = \frac{56.25 \text{ W}}{71.62 \text{ W}} \times 100\% \\ &= 78.54\% \text{ as expected} \end{aligned}$$

(d) Output voltage for maximum power operation is

$$V_{L(p)} = V_{CE(p)} = V_{CC} = 30 \text{ V}$$

Since Q_1 and Q_2 are emitter followers, the corresponding input voltage is

$$V_{i(p)} = V_{L(p)} = 30 \text{ V}$$

(e) Collector dissipation in both transistors at this input voltage is

$$\begin{aligned} P_{2Q} &= P_{i(dc)\max} - P_{o(ac)\max} \\ &= 71.62 \text{ W} - 56.25 \text{ W} = 15.37 \text{ W} \end{aligned}$$

Power dissipation in each transistor is

$$P_Q = \frac{P_{2Q}}{2} = \frac{15.37 \text{ W}}{2} = 7.685 \text{ W}$$

Example 7.40

For the circuit of example 7.39, calculate the maximum power dissipated by the output transistors and the input voltage at which this occurs. Comment on the result.

Solution

Given

$$V_{CC} = 30 \text{ V}$$

$$R'_L = R_L = 8 \Omega$$

From Equation (7.127), the maximum power dissipation in the output transistors is

$$P_{2Q(\max)} = \frac{2 V_{CC}^2}{\pi^2 R_L} = \frac{2 (30 \text{ V})^2}{\pi^2 (8 \Omega)} = 22.79 \text{ W}$$

Maximum power dissipation in each transistor is

$$P_{Q(\max)} = \frac{P_{2Q(\max)}}{2} = \frac{22.79 \text{ W}}{2} = 11.395 \text{ W}$$

From Equation (7.126), the corresponding output voltage is

$$V_{L(p)} = V_{CE(p)} = 0.636 V_{CC} = (0.636) (30 \text{ V}) = 19.08 \text{ V}$$

Due to emitter follower action the corresponding input voltage is

$$V_{i(p)} = V_{L(p)} = 19.08 \text{ V}$$

From the results of example 7.39, we find when $V_{i(p)} = 30 \text{ V}$, the ac power output is maximum and the corresponding total power dissipation is 15.37 W.

In the present example, we found that, the total power dissipation has a maximum value of 22.79 W when $V_{i(p)} = 19.08 \text{ V}$, which does not correspond to maximum power output.

Thus as stated earlier the maximum power dissipation in the output transistors does not occur when the circuit is delivering maximum power output but it occurs at an output power which is smaller than the maximum value.

◆ 7.7 CLASS D POWER AMPLIFIER

A class *D* power amplifier is designed to operate with digital or pulse-type signals. It has an efficiency of more than 90% which is highly desirable for a power amplifier. The name class *D* arises from the fact that, the circuit is designed to operate with digital or pulse-type signals.

Figure 7.24 shows the block diagram of class *D* power amplifier. The sinusoidal input signal V_i is compared with saw tooth waveform in a comparator. The comparator output switches from low to high level whenever the voltage level of sawtooth waveform goes above the voltage level of the input waveform. The comparator output switches from high to low level when the sawtooth voltage level falls below the input voltage level. Thus the output of comparator is a digital waveform as shown in Fig. 7.25.

The high output of comparator turns on the amplifiers transistor devices which are usually power BJTs or power MOSFETs. The transistors are turned off by the low output of comparator. The transistors provide output current only when they are turned on with little power loss due to their low on-state voltage. Once again the output of the amplifier is a pulse-type signal. The low-pass filter converts the pulse-type signal back to sinusoidal signal. It is important to note that the output signal is a large amplitude sinusoid whose frequency is same as the input sinusoid. The circuit operates with veryhigh efficiency since most of the power applied to the amplifier is transferred to the load. A small amount of feedback is applied to improve the linearity of the circuit.

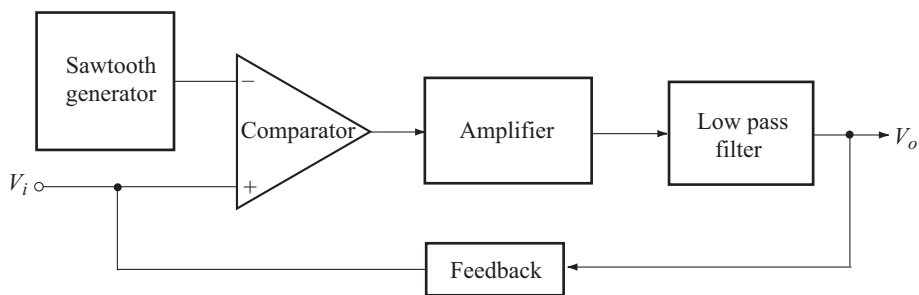


Fig. 7.24 Block diagram of class D power amplifier

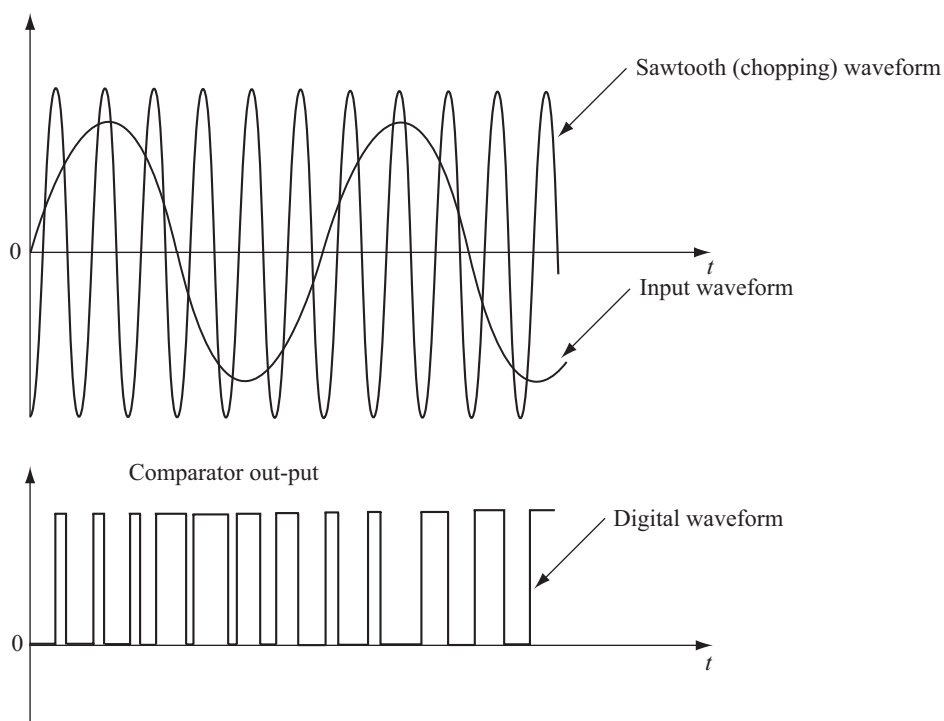


Fig. 7.25 Waveforms in class-D power amplifier

◆ 7.8 POWER TRANSISTOR HEAT SINKING

Due to power dissipation in the power transistor, the temperature of the collector-base junction rises. If the junction temperature exceeds the maximum permissible value, the transistor will get destroyed due to thermal runaway. Thus the temperature of the collector-base junction places an upper limit on the allowable power dissipation. Typical maximum junction temperature for silicon transistors is $150\text{--}200^\circ\text{C}$ and for germanium transistor, it is $100\text{--}110^\circ\text{C}$. Note that the power dissipation capability of silicon transistors is higher than that of germanium transistor.

The average power dissipated in a transistor may be approximated by

$$P_D = V_{CE} I_C \quad (7.129)$$

It is important to note that, the power dissipation capacity of a transistor decreases with increase in junction temperature. This is called power derating. For instance, as specified in the data sheets, the maximum power dissipation of the transistor 2N1936 is 4 W at 25°C and it decreases to zero at a junction temperature of 175°C.

One way to increase the power rating of a transistor is to get rid of the heat faster. Heat sinks are used for this purpose. With heat sinks, the surface area of the transistor case increases which helps the heat to escape more easily into the surrounding air.

Figure 7.26(a) shows the push-on heat sink. When the heat sink is pushed on to the transistor case, heat radiates more quickly because of the increased surface area of the fins.

Figure 7.26(b) shows the power-tab transistor. The metal tab provides a path out of the transistor for heat. This metal tab can be fastened to the chassis of electronic equipment. Because the chassis is a massive heat sink, heat can easily escape from the transistor to the chassis.

In large power transistors, the collector is connected directly to the case as shown in Fig. 7.26(c) to allow the heat escape as easily as possible. The transistor case is then fastened to the chassis. To prevent the collector from shorting to the chassis ground, a thin insulating washer and a thermal conductive paste are used between the transistor case and the chassis.

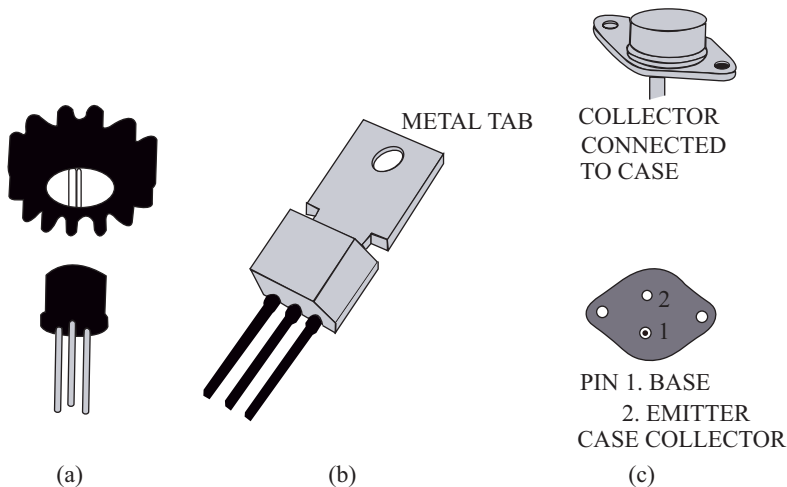


Fig. 7.26 (a) Push-on heat sink
(b) Power-tab transistor
(c) Power transistor with collector connected to case

Figure 7.27 shows a typical power derating curve for a silicon transistor. The curve shows that the power derating takes place linearly after a certain temperature specified by the manufacturer. Note that for silicon transistor the power dissipation reduces to 0 W at a case temperature of 200°C.

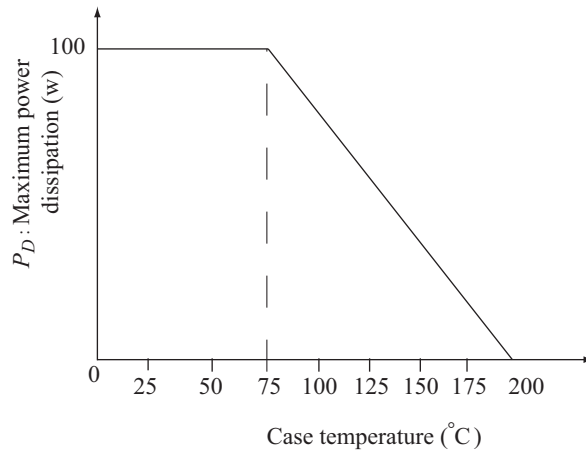


Fig. 7.27 Typical power derating curve for silicon transistor

The slope of power derating curve is called the derating factor denoted by, D , expressed in watts or milli watts per degree centigrade of temperature.

$$D = \frac{\text{Decrease in power rating}}{\text{Increase in case temperature}}$$

$$D = \frac{P_{D_o} - P_{D_1}}{T_1 - T_o} \quad (7.130)$$

where P_{D_o} = Maximum power dissipation at temperature T_o

P_{D_1} = Maximum power dissipation at temperature T_1

For instance, From Fig. 7.27 we find that when the case temperature rises from 75 °C to 200 °C, the maximum power dissipation decreases from 100 W to 0 W.

$$\therefore D = \frac{100 \text{ W}}{200^\circ\text{C} - 75^\circ\text{C}} = 0.8 \text{ W}/^\circ\text{C}$$

Example 7.41

A silicon transistor is rated for 100 W at 25 °C. Calculate the maximum dissipation at a case temperature of 125° C if the derating is required above 25° C with a derating factor of 0.5 W/°C.

Solution

$$P_{D_o} = 100 \text{ W} \quad T_o = 25^\circ\text{C}$$

$$T_1 = 125^\circ\text{C} \quad D = 0.5 \text{ W}/^\circ\text{C}$$

$$D = \frac{P_{D_o} - P_{D_1}}{T_1 - T_o}$$

$$0.5 \text{ W}/^\circ\text{C} = \frac{100 \text{ W} - P_{D_1}}{125^\circ\text{C} - 25^\circ\text{C}}$$

$$\begin{aligned}
 P_{D1} &= 100 \text{ W} - (100^\circ\text{C}) (0.5 \text{ W}/^\circ\text{C}) \\
 &= 50 \text{ W}
 \end{aligned}$$

Example 7.42

A silicon transistor is rated for 100 W at 25 °C. If the derating factor is 0.6 W/°C, calculate the maximum power dissipation at a case temperature of boiling temperature of water.

Solution

$$\begin{aligned}
 P_{D_o} &= 100 \text{ W} & T_o &= 25^\circ\text{C} \\
 D &= 0.6 \text{ W}/^\circ\text{C} & T_1 &= 100^\circ\text{C} \text{ (Boiling temperature of water)} \\
 D &= \frac{P_{D_o} - P_{D_1}}{T_1 - T_o} \\
 0.6 \text{ W}/^\circ\text{C} &= \frac{100 \text{ W} - P_1}{100^\circ\text{C} - 25^\circ\text{C}} \\
 P_1 &= 100 \text{ W} - (75^\circ\text{C}) (0.6 \text{ W}/^\circ\text{C}) = 55 \text{ W}
 \end{aligned}$$

◆ 7.9 THERMAL ANALOGY OF A POWER TRANSISTOR

The heat produced at the collector-base junction passes through the transistor case to the heat sink and then radiates in to the surrounding air. The temperature of this air is known as the ambient temperature which is around 25 °C but it can be much higher on hot days.

Let T_J = Junction temperature
 T_C = Case temperature
 T_H = Heat sink temperature.
 T_A = Ambient temperature.

All these temperatures are measured with respect to absolute zero.

If is found experimentally that the steady-state temperature rise at the collector junction is proportional to the power dissipated at the junction.

$$\left. \begin{aligned}
 \text{ie} & \quad T_J - T_A \propto P_D \\
 \text{or} & \quad T_J - T_A = \theta_{JA} P_D
 \end{aligned} \right\} \quad (7.131)$$

$$\Rightarrow \quad \theta_{JA} = \frac{T_J - T_A}{P_D} \quad (7.132)$$

θ_{JA} is the thermal resistance from junction to ambient (total thermal resistance). Since heat flows from junction to case, case to heat sink and then from heat sink to ambient, using electrical analogy of thermal resistances, we can write

$$\theta_{JA} = \theta_{JC} + \theta_{CS} + \theta_{SA} \quad (7.133)$$

where θ_{JC} = transistor thermal resistance (junction to case)
 θ_{CS} = Insulator thermal resistance (case to heat sink)
 θ_{SA} = heat-sink thermal resistance (heat sink to ambient)

Figure 7.28 Shows the electrical analogous circuit using the thermal resistances.

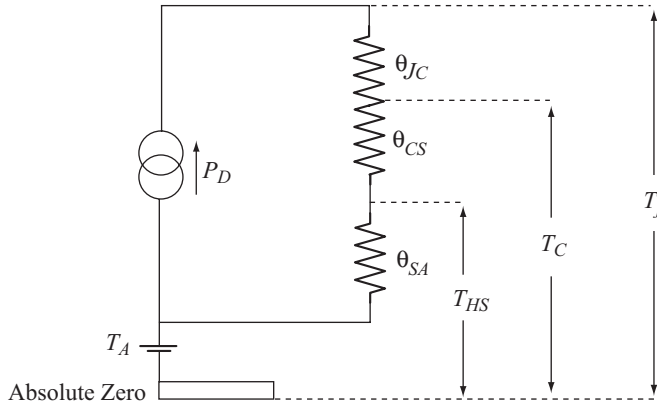


Fig. 7.28 Thermal-to-electrical analogy

From Equation (7.132)

$$T_j = P_D \theta_{JA} + T_A \quad (7.134)$$

Note that the junction temperature floats on the ambient temperature. Also higher the ambient temperature lower is the allowed value of device power dissipation. The following example illustrates how the heat sink increases the power handling capacity of the device for a given rise in junction temperature.

Example 7.43

A power transistor using heat sink is required to dissipate 50 W. The thermal resistance values are as follows.

$$\begin{aligned} \theta_{SA} &= 2^\circ \text{C/W} \\ \theta_{CS} &= 0.8^\circ \text{C/W} \\ \theta_{JC} &= 0.5^\circ \text{C/W} \end{aligned}$$

- Calculate the rise in Junction temperature.
- If $\theta_{JA} = 40^\circ \text{C/W}$ without heat sink, what would be the rise in Junction temperature. Comment on the result.

Solution(a) *With heat sink*

$$\begin{aligned}\theta_{JA} &= \theta_{JC} + \theta_{CS} + \theta_{SA} \\ &= 0.5^\circ \text{C/W} + 0.8^\circ \text{C/W} + 2^\circ \text{C/W} = 3.3^\circ \text{C/W}\end{aligned}$$

Rise in Junction temperature is

$$T_J - T_A = P_D \theta_{JA} = (50 \text{ W}) (3.3^\circ \text{C/W}) = 165^\circ \text{C}$$

(b) *Without heat sink*

$$\begin{aligned}\theta_{JA} &= 40^\circ \text{C/W} \\ T_J - T_A &= (50 \text{ W}) (40^\circ \text{C/W}) = 2000^\circ \text{C} !!!\end{aligned}$$

This excessive high temperature will certainly destroy the transistor.

Example 7.44

A silicon power transistor operating with heat sink has the following data.

Transistor rating: : 150 W at 25 °C

Transistor thermal resistance, θ_{JC} : 0.5°C/WInsulator thermal resistance, θ_{CS} : 0.6° C/WHeat-sink thermal resistance, θ_{SA} : 1.5° C/WWhat maximum power can be dissipated if the ambient temperature is 40°C and $T_{j\max} = 200^\circ \text{C}$?**Solution**

$$\begin{aligned}\theta_{JA} &= \theta_{JC} + \theta_{CS} + \theta_{SA} \\ &= 0.5^\circ \text{C/W} + 0.6^\circ \text{C/W} + 1.5^\circ \text{C/W} = 2.6^\circ \text{C/W} \\ P_D &= \frac{T_J - T_A}{\theta_{JA}} = \frac{200^\circ \text{C} - 40^\circ \text{C}}{2.6^\circ \text{C/W}} = 61.54 \text{ W}\end{aligned}$$

Example 7.45What maximum power can a silicon transistor dissipate into free air at an ambient temperature of 80° C. The junction temperature should not exceed 200° C. Take θ_{JA} 40° C/W.**Solution**

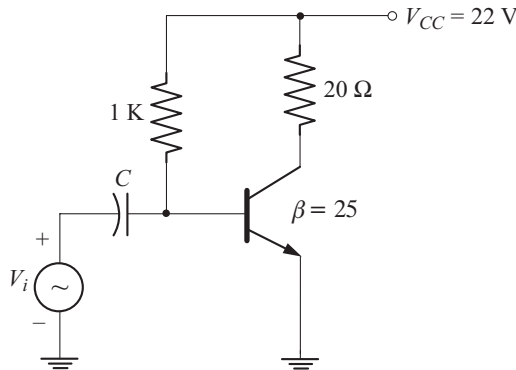
In this case the transistor operates without heat sink.

$$\begin{aligned}P_D &= \frac{T_J - T_A}{\theta_{JA}} \\ &= \frac{200^\circ \text{C} - 80^\circ \text{C}}{40^\circ \text{C/W}} \\ &= 3 \text{ W}\end{aligned}$$



EXERCISE PROBLEMS

- 7.1 For the class A series-fed power amplifier shown below, calculate the dc input power, ac output power and efficiency. The input voltage v_i results in a base current of 7.07 mA(rms).



- 7.2 The following data is available for a class A series-fed Power amplifier.

$$V_{CE\max} = 14 \text{ V}, V_{CE\min} = 6 \text{ V}, I_{C\max} = 7 \text{ mA}, I_{C\min} = 3 \text{ mA}$$

- calculate the ac power delivered to the load.
 - conversion efficiency if $V_{CC} = 10 \text{ V}$ and $I_C = 5 \text{ mA}$.
- 7.3 What transformer turns ratio is required to match a 16Ω speaker load so that the effective load resistance seen at the primary is 10 K ?
- 7.4 A class B Power amplifier is delivering an output voltage of 10 V peak to a 8Ω load. If the dc power supply is 30 V , calculate
- dc power input,
 - ac power delivered to the load.
 - conversion efficiency and
 - power dissipated in the collector of each transistor.
- 7.5 A class B Power amplifier is driving a load at 16Ω . If the supply voltage $V_{CC} = 25 \text{ V}$, calculate
- maximum ac power output
 - maximum dc power input
 - collector dissipation in each transistor.
- 7.6 A class B power amplifier has a dc supply voltage of $V_{CC} = 30 \text{ V}$. Calculate the efficiency when the output voltage is
- 30 V peak
 - 15 V peak
 - 7.5 V peak
- 7.7 The following readings are obtained for a Power amplifier during measurement of distortion.

$$V_{CE\min} = 3 \text{ V}, V_{CE\max} = 25 \text{ V}, V_{CEQ} = 15 \text{ V}$$

Calculate the second harmonic distortion.

Chapter 8

OSCILLATORS

Oscillators are an important class of circuits and are used in almost every electronic systems. For example, oscillators are employed to produce sinusoidal signals that are used as carrier in radio and television broadcasts. Oscillators are also used to produce the square wave used as clocks in computers and other synchronous digital systems. This chapter covers the basic principle of operation and design aspects of RC, LC and crystal oscillators.

◆ 8.1 BASIC PRINCIPLE OF OSCILLATORS

An oscillator is a circuit designed to provide periodic output with no input signal. It requires only the DC power. An oscillator is basically an amplifier with positive feedback.

Fig 8.1 shows an amplifier, a feedback network and an inverting network not yet connected to form a closed loop.

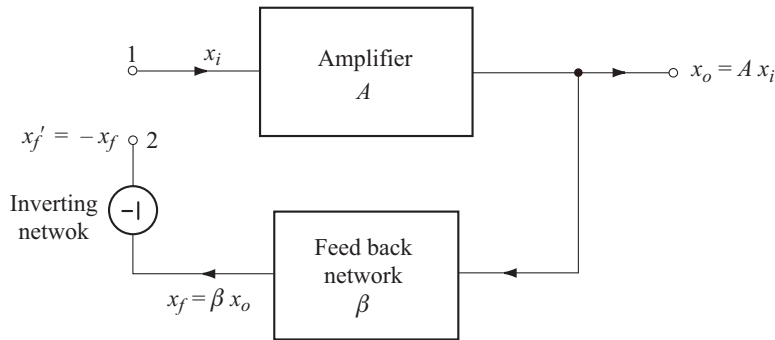


Fig. 8.1 Illustration of basic principle of oscillators

The amplifier provides an output signal x_o in response to the signal x_i applied directly to the amplifier input terminal. The output of the feedback network is $x_f = \beta x_o = A \beta x_i$ and the output of the inverting network is $x_f' = -x_f = -A \beta x_i$. Now

$$\text{Loop gain} = \frac{x_f'}{x_i} = \frac{-A \beta x_i}{x_i} = -A \beta \quad (8.1)$$